

Retail Data Engineering Project

Azure SQL Database | Azure Data Factory | Azure Data Lake Storage | Azure Databricks | Power BI

Prepared by Nandini Babu | September 2026

Project overview

This project moves retail source data into an analytical dashboard. Azure Data Factory copies products, stores and transactions from Azure SQL Database and customers from an HTTP JSON source into bronze storage. Azure Databricks reads the landed files, cleans them into silver tables, computes sales and daily store totals in gold, and exposes the results through a SQL warehouse for Power BI. The screenshots document the configuration, loaded tables, transformations and final dashboard.

Verified result

A Databricks query against catalogretail.gold.sales shows 30 transactions, 101 units sold and total sales of 134,570.74. The Power BI screenshots show a report with sales by store, a trend, a category view and KPI visuals. These figures describe the captured project state; they are not a separate rerun of the cloud environment.

Source and destination map

Stage	Objects shown in the project
Azure SQL source	dbo.products, dbo.stores, dbo.transactions
HTTP source	customers JSON through LS_Customers_HTTP
ADF landing	PL_Retail_To_Bronze and bronze datasets
ADLS storage	bronze, silver and gold containers
Databricks	catalogretail silver and gold tables
Presentation	Databricks SQL warehouse to Power BI

Evidence note: The SQL creation screenshot was not captured during execution. The following labeled example illustrates a possible table definition from the visible preview columns. It is not presented as a screenshot from the Azure account or a copy of the exact script used.

1 Storage containers and source tables

The storage account contains bronze, silver and gold containers. Bronze holds data landed from source systems; silver stores cleaned datasets; gold stores business ready results. The last storage screenshot shows gold sales and daily store sales outputs after Databricks processing.

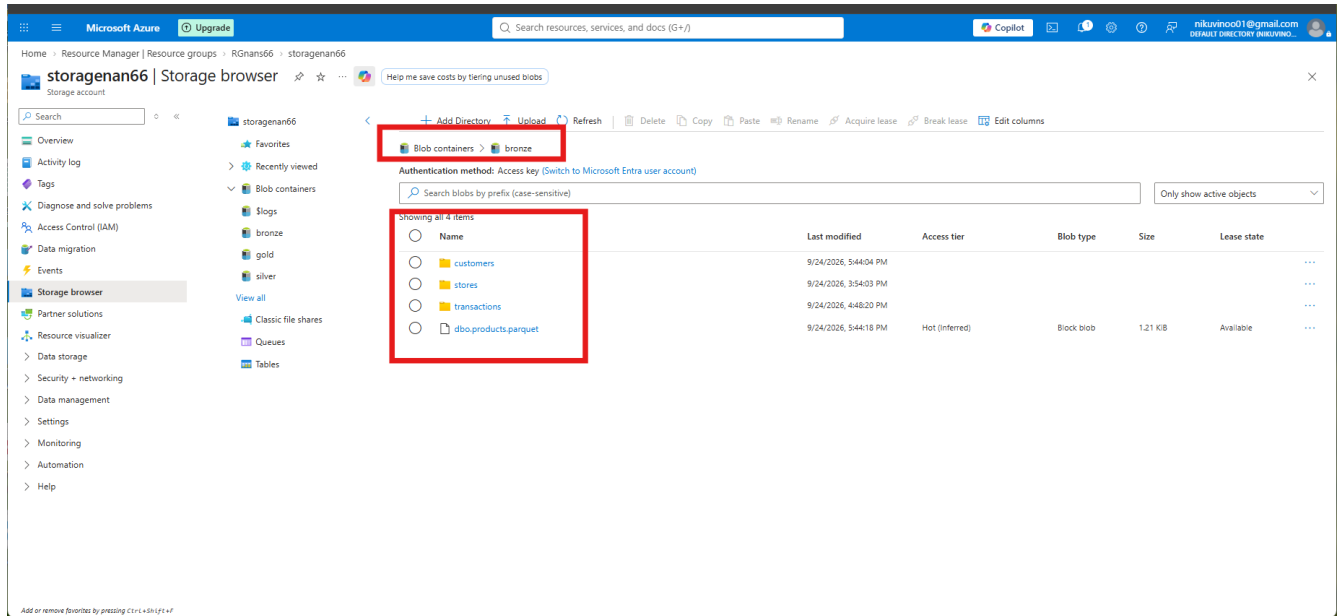


Figure 1. Bronze container with landed source data

The bronze view shows customers, stores, transactions and a products Parquet object.

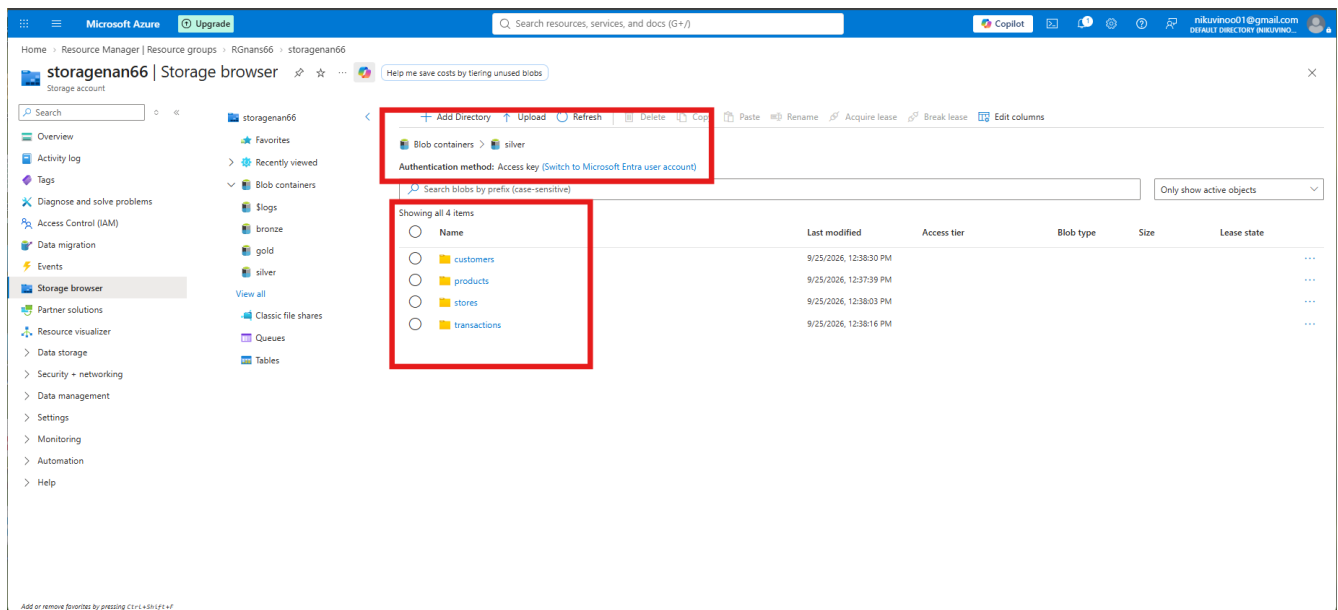


Figure 2. Silver container with cleaned datasets

Customers, products, stores and transactions appear under silver.

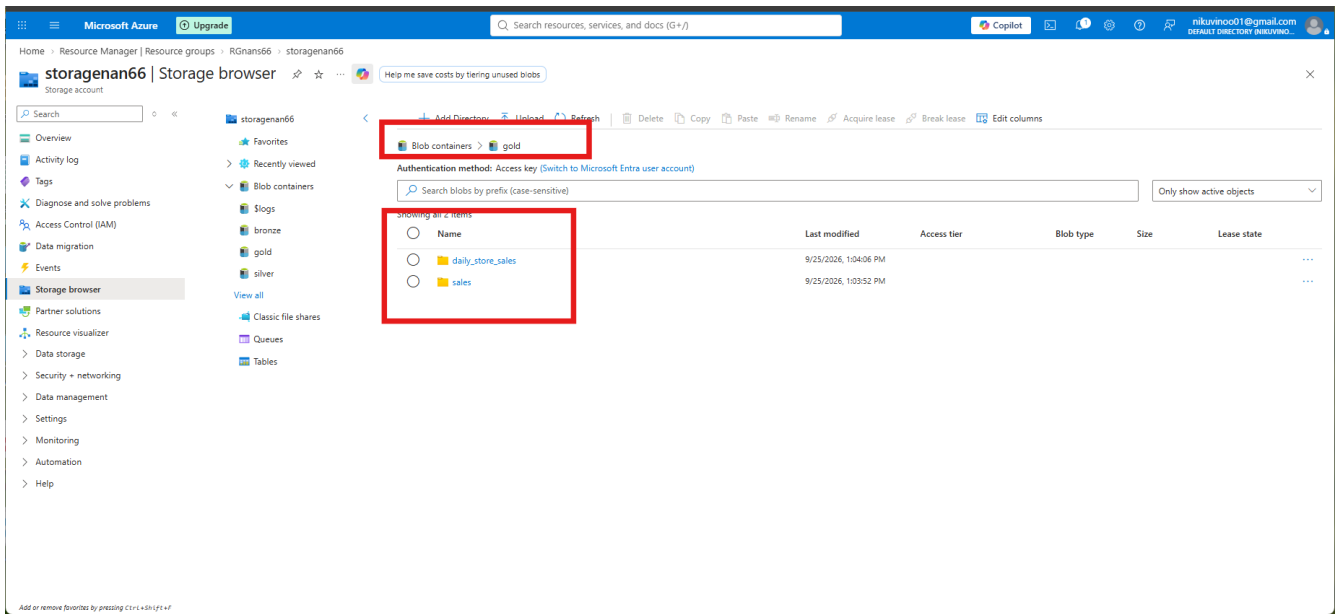


Figure 3. Gold container with analytical outputs

The gold container contains sales and daily store sales outputs.

2 Azure SQL table creation

Three SQL tables supply the structured retail records. The supplied ADF previews show products with product ID, name, category and price; stores with store ID, name and location; and transactions with transaction ID, customer ID, product ID, store ID, quantity and transaction date. Customers come from the HTTP JSON source.

SQL table creation | reconstructed example

Illustrative Azure SQL schema based on the columns visible in the supplied previews

```
CREATE TABLE dbo.products (  
    product_id INT PRIMARY KEY,  
    product_name NVARCHAR(150),  
    category NVARCHAR(80),  
    price DECIMAL(12,2)  
);  
  
CREATE TABLE dbo.stores (  
    store_id INT PRIMARY KEY,  
    store_name NVARCHAR(120),  
    location NVARCHAR(100)  
);  
  
CREATE TABLE dbo.transactions (  
    transaction_id INT PRIMARY KEY,  
    customer_id INT,  
    product_id INT,  
    store_id INT,  
    quantity INT,  
    transaction_date DATE  
);
```

Not an original Azure SQL screenshot or an execution record

Reconstructed SQL table creation example. This is an illustrative code image based on observed columns, not the original SQL editor screenshot or verified DDL. Primary keys, data types and lengths should be compared with the actual database before reuse.

3 Source data verification in Azure Data Factory

The ADF dataset previews confirm that SQL source tables were accessible through the Azure SQL linked service. The examples below show the rows and column names that support the later transformations.

The screenshot shows the Azure Data Factory interface. In the 'Factory Resources' pane on the left, the 'Datasets' section is expanded, and 'DS_SQL_Products' is highlighted with a red box. The 'Preview data' pane in the center shows the 'dbo.products' dataset from the 'LS_Retail_SQL' linked service. The table has the following data:

	product_id	product_name	category	price
1	1	Wireless Mouse	Electronics	799.99
2	2	Bluetooth Speaker	Electronics	1299.49
3	3	Yoga Mat	Fitness	499.00
4	4	Laptop Stand	Accessories	999.99
5	5	Notebook Set	Stationery	149.00
6	6	Water Bottle	Fitness	299.00
7	7	Smartwatch	Electronics	4999.00
8	8	Desk Organizer	Accessories	399.00
9	9	Dumbbell Set	Fitness	1999.00
10	10	Pen Drive 32GB	Electronics	599.00

The 'Properties' pane on the right shows the dataset name 'DS_SQL_Products' and a description field.

Figure 4. Preview of dbo.products

Visible fields include product_id, product_name, category and price.

The screenshot shows the Azure Data Factory interface. In the 'Factory Resources' pane on the left, the 'Datasets' section is expanded, and 'DS_SQL_Stores' is highlighted with a red box. The 'Preview data' pane in the center shows the 'dbo.stores' dataset from the 'LS_Retail_SQL' linked service. The table has the following data:

	store_id	store_name	location
1	1	City Mall Store	Mumbai
2	2	High Street Store	Delhi
3	3	Tech World Outlet	Bangalore
4	4	Downtown Mini Store	Pune
5	5	Mega Plaza	Chennai

The 'Properties' pane on the right shows the dataset name 'DS_SQL_Stores' and a description field. At the bottom, the 'Partition option' is set to 'None'.

Figure 5. Preview of dbo.stores

The store master contains store_id, store_name and location.

Microsoft Azure | Data Factory | adfman66

Upgrade your Azure Data Factory to the native Microsoft Fabric

Factory Resources

- Pipelines 1
 - PI_Retail_To_Bronze
- Change Data Capture (preview) 0
- Datasets 6
 - DS_Bronze_Products
 - DS_Bronze_Stores
 - DS_Bronze_Transactions
 - DS_SQL_Products
 - DS_SQL_Stores
 - DS_SQL_Transactions
- Data flows 0
- Power Query 0

Preview data

Linked service: LS_Retail_SQL

Object: dbo.transactions

	transaction_id	customer_id	product_id	store_id	quantity	transaction_date
1	1	127	8	4	4	03/31/2025
2	2	105	3	4	5	11/12/2024
3	3	116	2	2	3	05/01/2025
4	4	120	8	1	1	11/02/2024
5	5	105	5	2	1	03/17/2025
6	6	110	7	3	5	01/04/2025
7	7	110	7	2	5	01/01/2025
8	8	126	7	5	2	06/08/2025
9	9	123	1	3	2	10/08/2024
10	10	124	2	2	5	08/27/2024

Properties

General Related (1)

Name DS_SQL_Transactions

Description

Annotations

+ New

Cancel

Figure 6. Preview of dbo.transactions

Visible fields include transaction_id, customer_id, product_id, store_id, quantity and transaction_date.

4 ADF connection and ingestion setup

Linked services connect the factory to Azure SQL, ADLS Gen2 and the customers HTTP source. Separate source and bronze datasets let the copy activities select each table and destination. The pipeline PL_Retail_To_Bronze contains copy activities for the retail datasets.

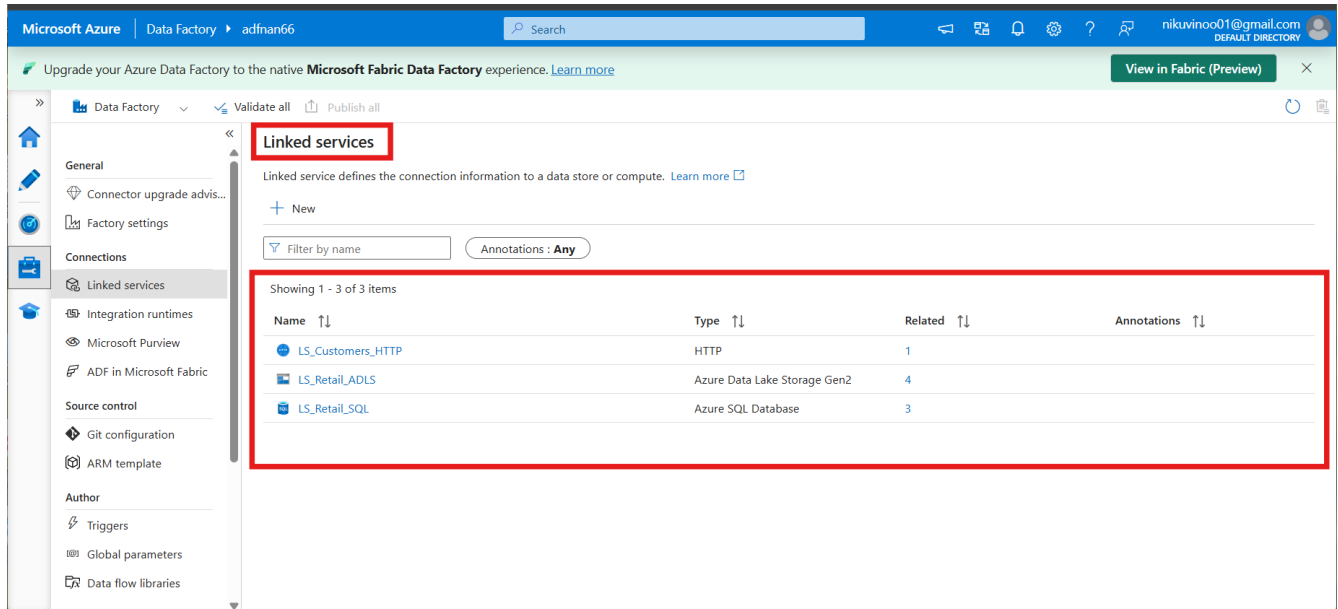


Figure 7. ADF linked services

The factory lists LS_Customers_HTTP, LS_Retail_ADLS and LS_Retail_SQL.

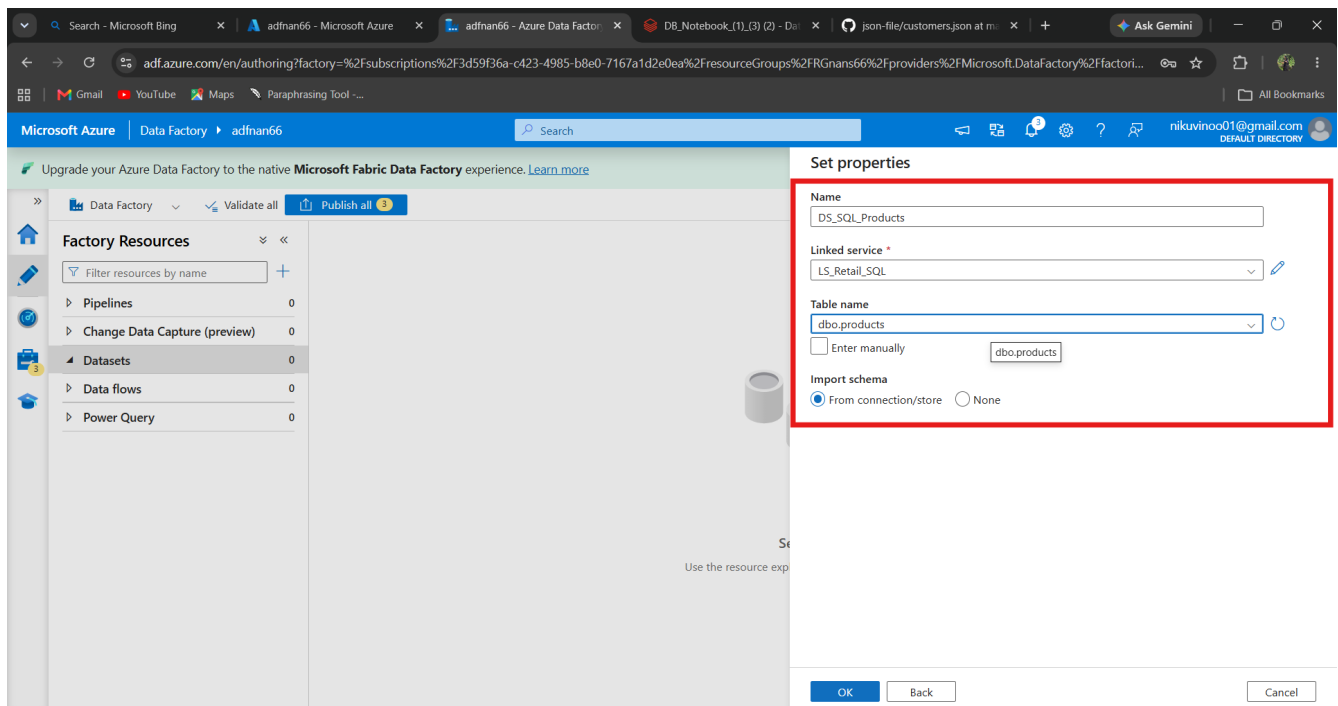


Figure 8. SQL Products dataset selection

The SQL source dataset is pointed to dbo.products.

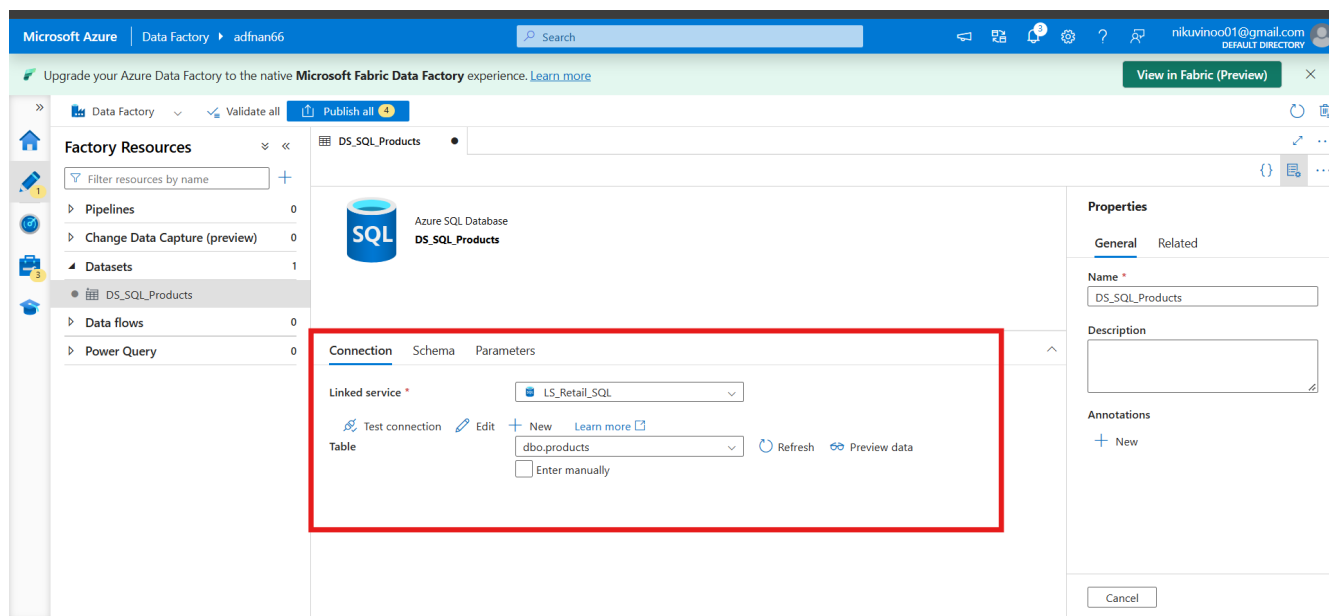


Figure 9. SQL Products dataset connection

The connection settings use the SQL linked service.

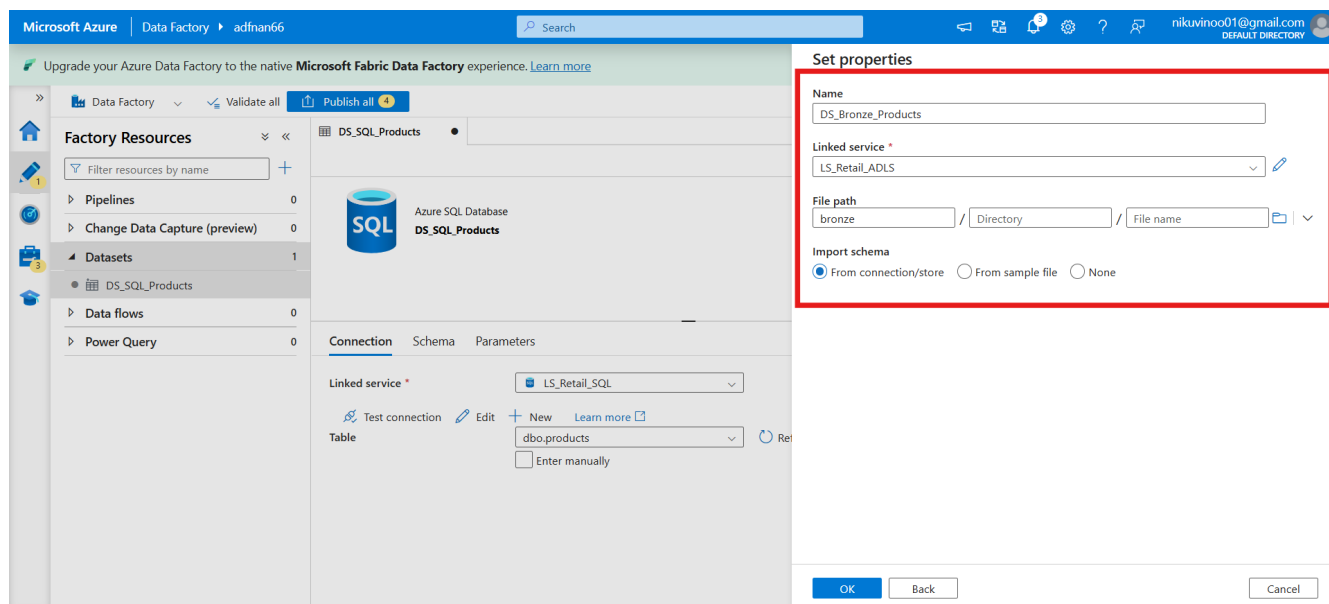


Figure 10. Bronze Products destination dataset

The dataset targets the bronze container through the ADLS linked service.

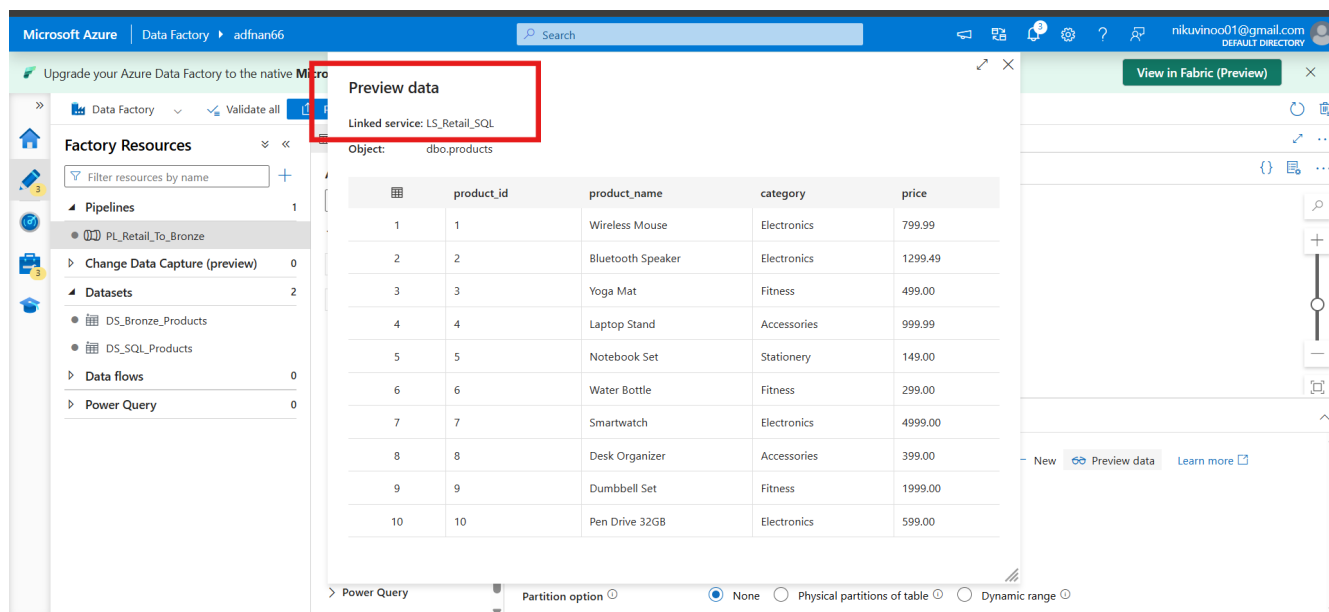


Figure 11. Products preview before copying

Rows are visible in the products source preview.

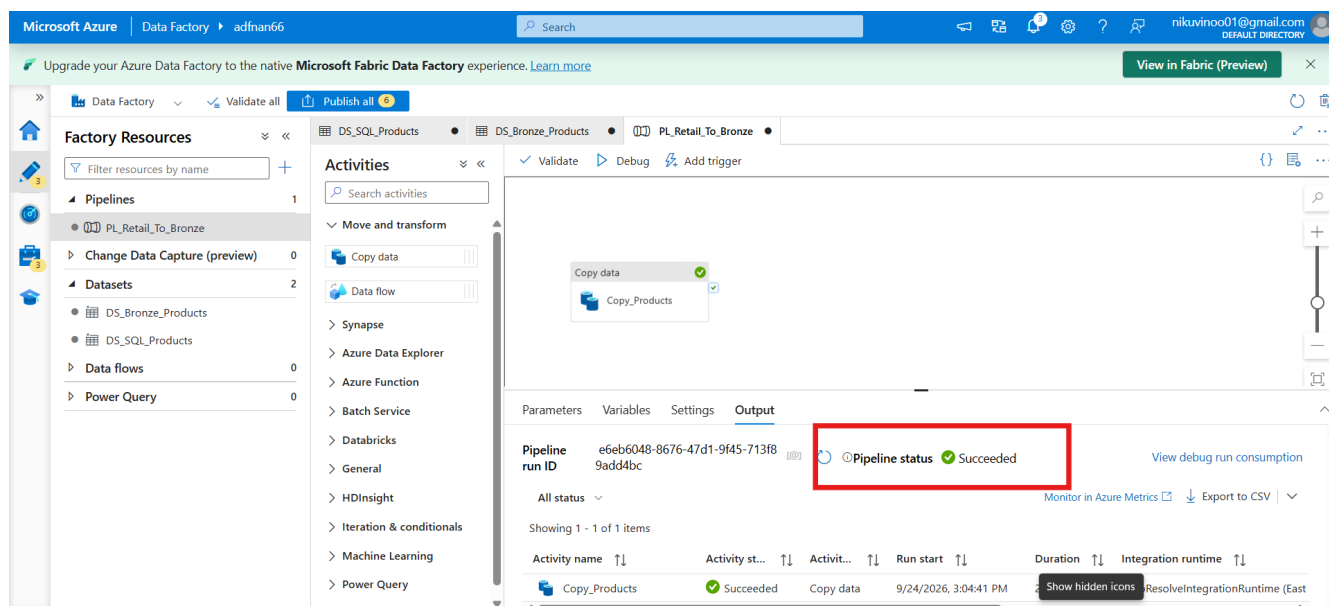


Figure 12. Products copy activity

The first copy activity is present in the retail to bronze pipeline.

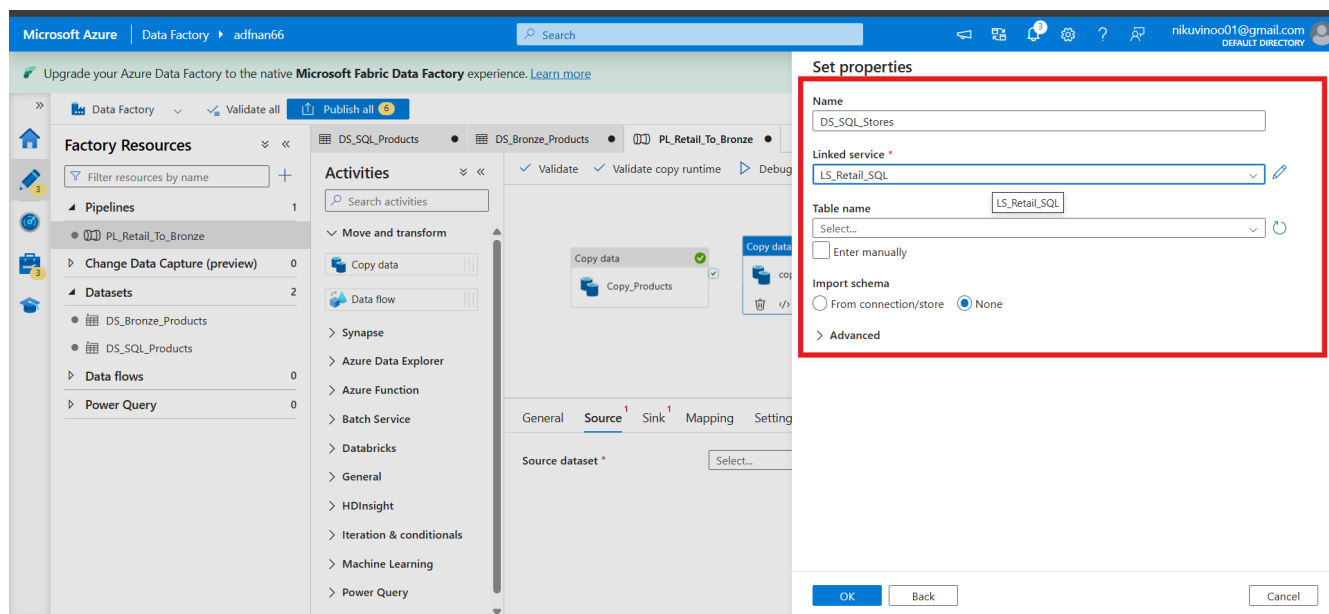


Figure 13. Stores source dataset configuration

A SQL source dataset is added for stores.

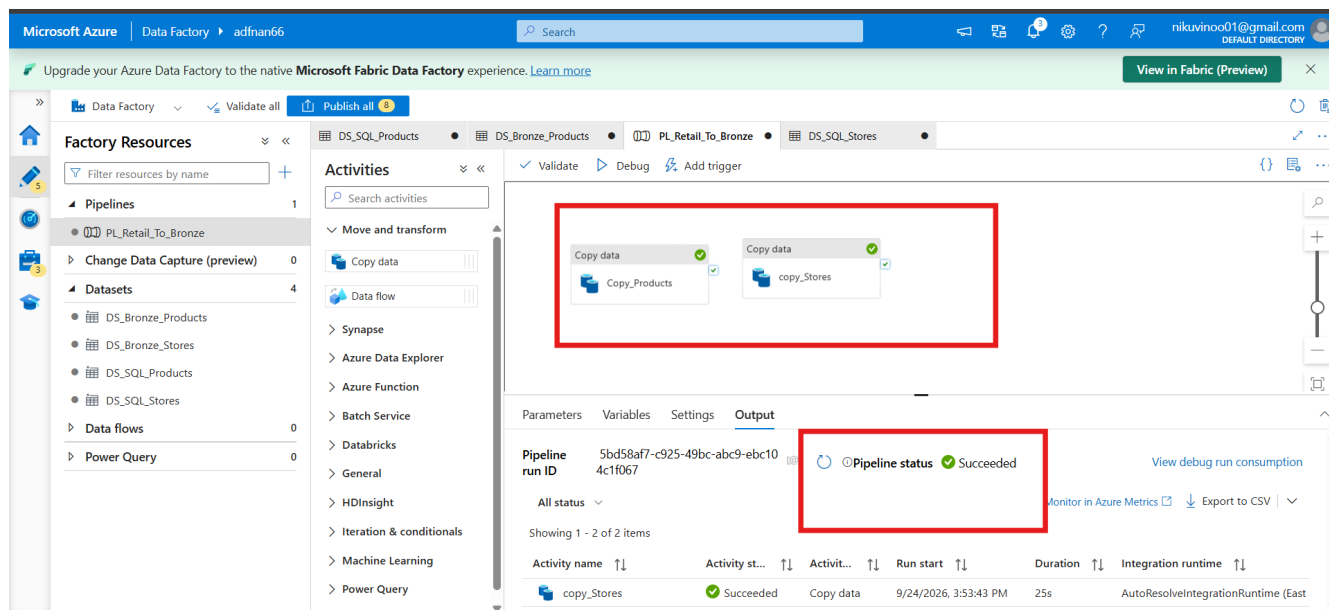


Figure 14. Stores copy activity

The pipeline now contains products and stores copies.

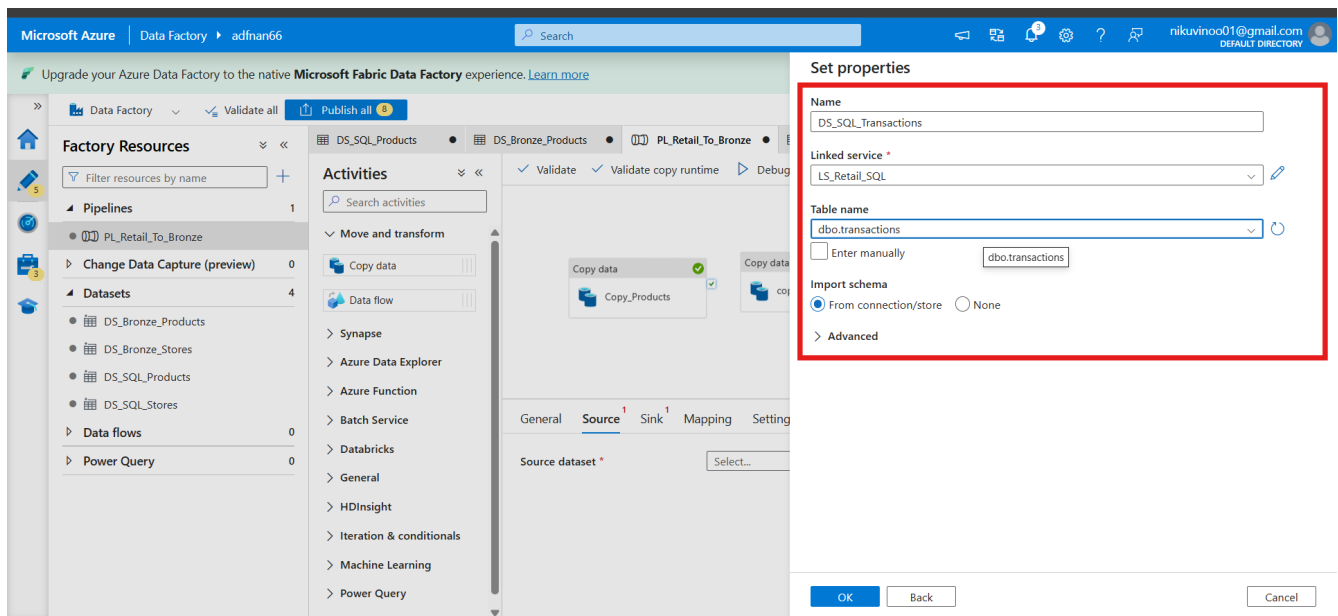


Figure 15. Transactions source dataset

The dataset selects dbo.transactions.

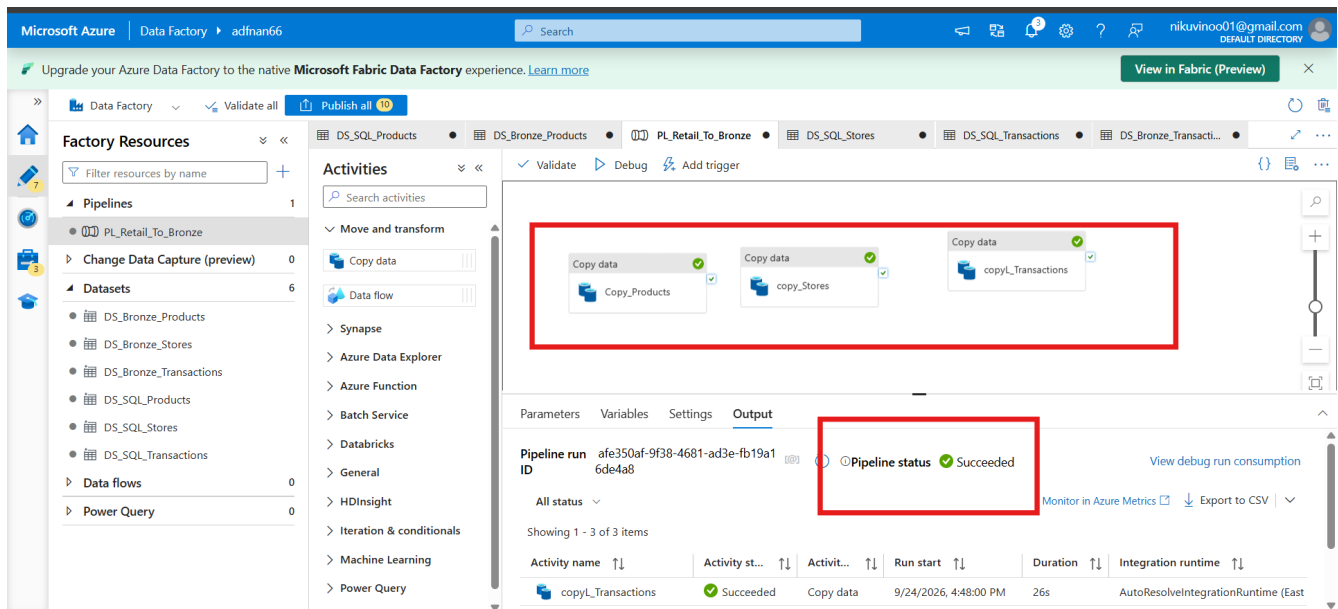


Figure 16. Transactions copy activity

Transactions are added to the pipeline.

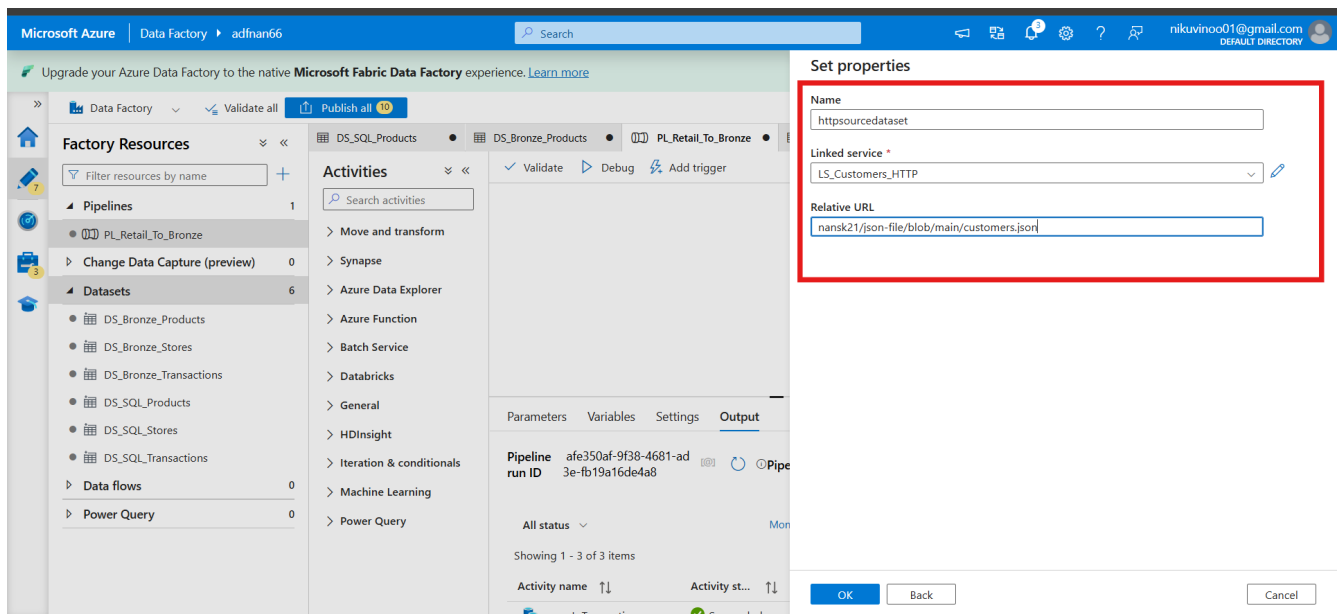


Figure 17. Customers HTTP source

The customers JSON endpoint is configured as an HTTP dataset.

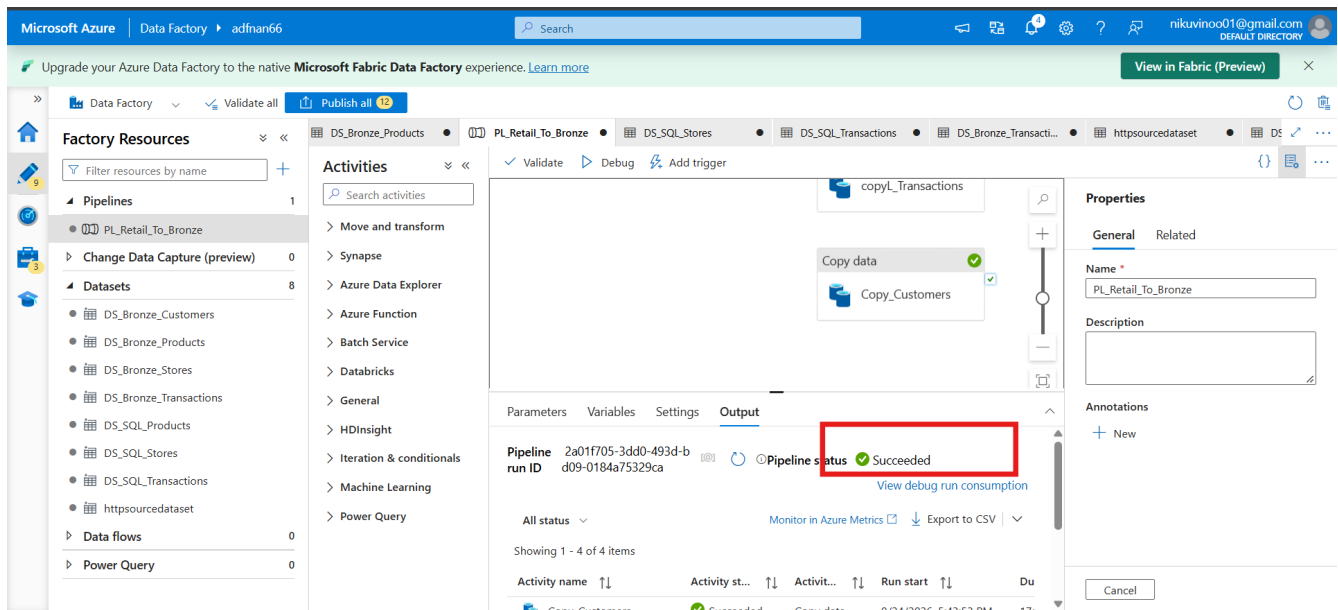


Figure 18. Customer ingestion activity

The source is integrated into the same retail pipeline.

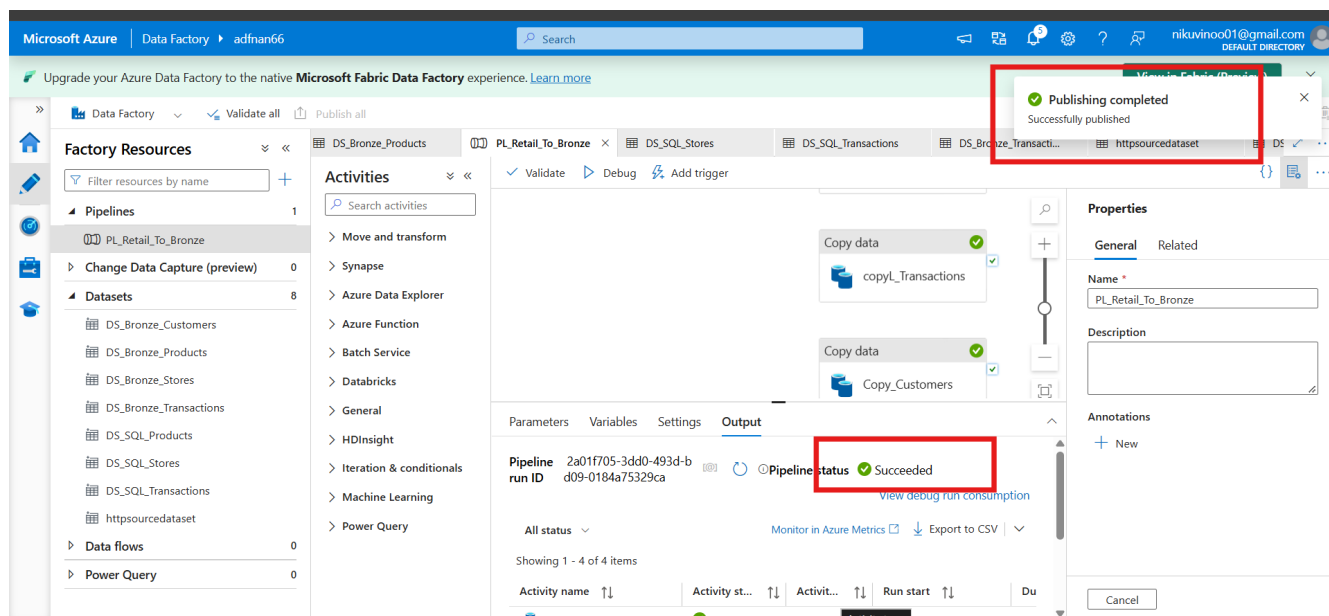


Figure 19. ADF publishing confirmation

The screenshot shows a successfully published factory.

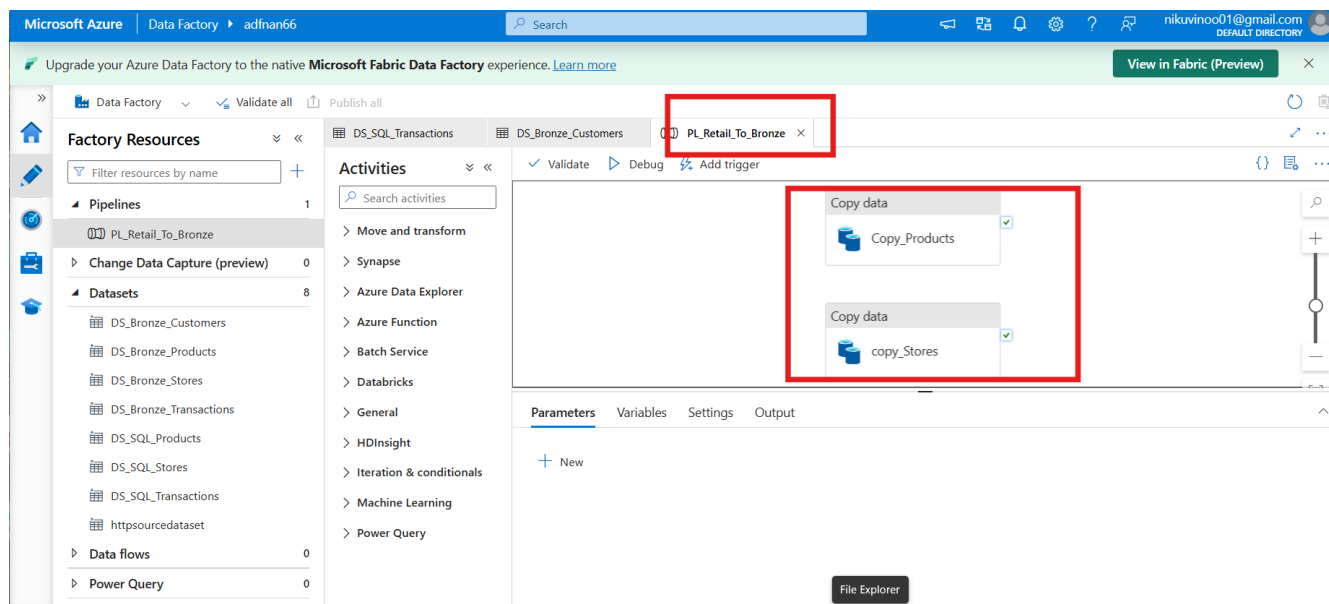


Figure 20. Final pipeline and datasets

The resource pane shows SQL and bronze datasets and the retail pipeline.

5 Databricks bronze silver and gold

The Databricks notebook reads files from bronze, transforms records into silver datasets, and builds sales aggregates in gold. The catalog shows four silver tables: customers, products, stores and transactions. It also shows two gold tables: sales and daily_store_sales. The selected screenshots capture transformations and a final query, but do not establish that every step is scheduled for automatic reruns.

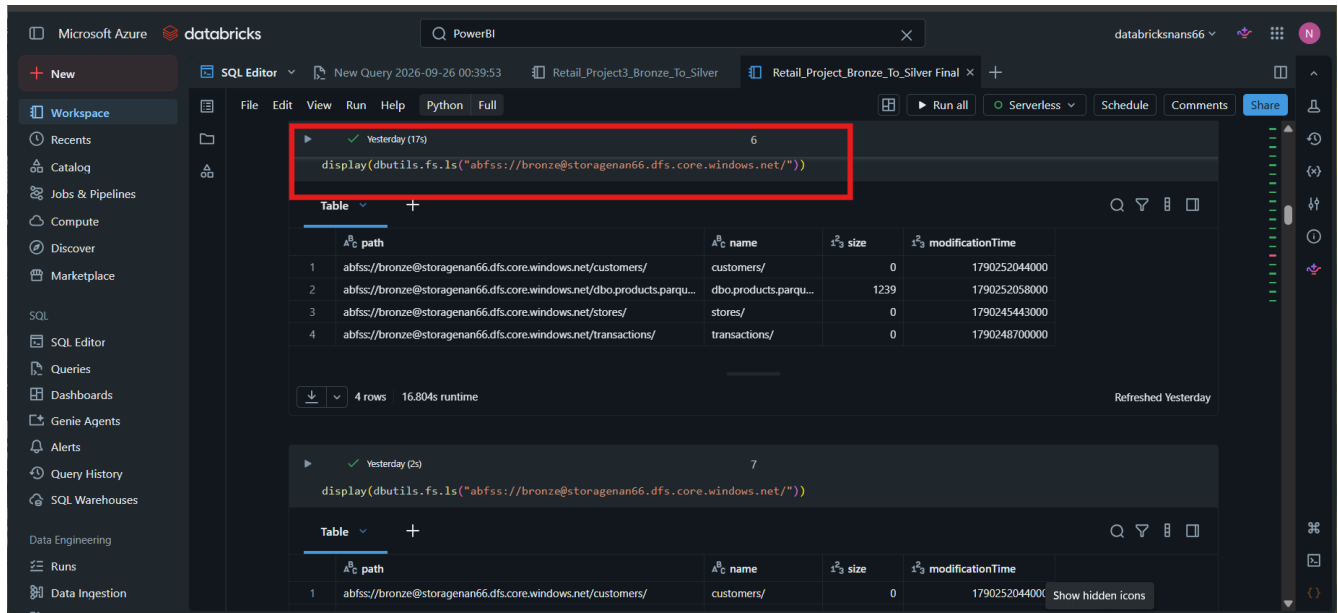


Figure 21. Bronze source inventory

The notebook lists customers, products, stores and transactions in bronze.

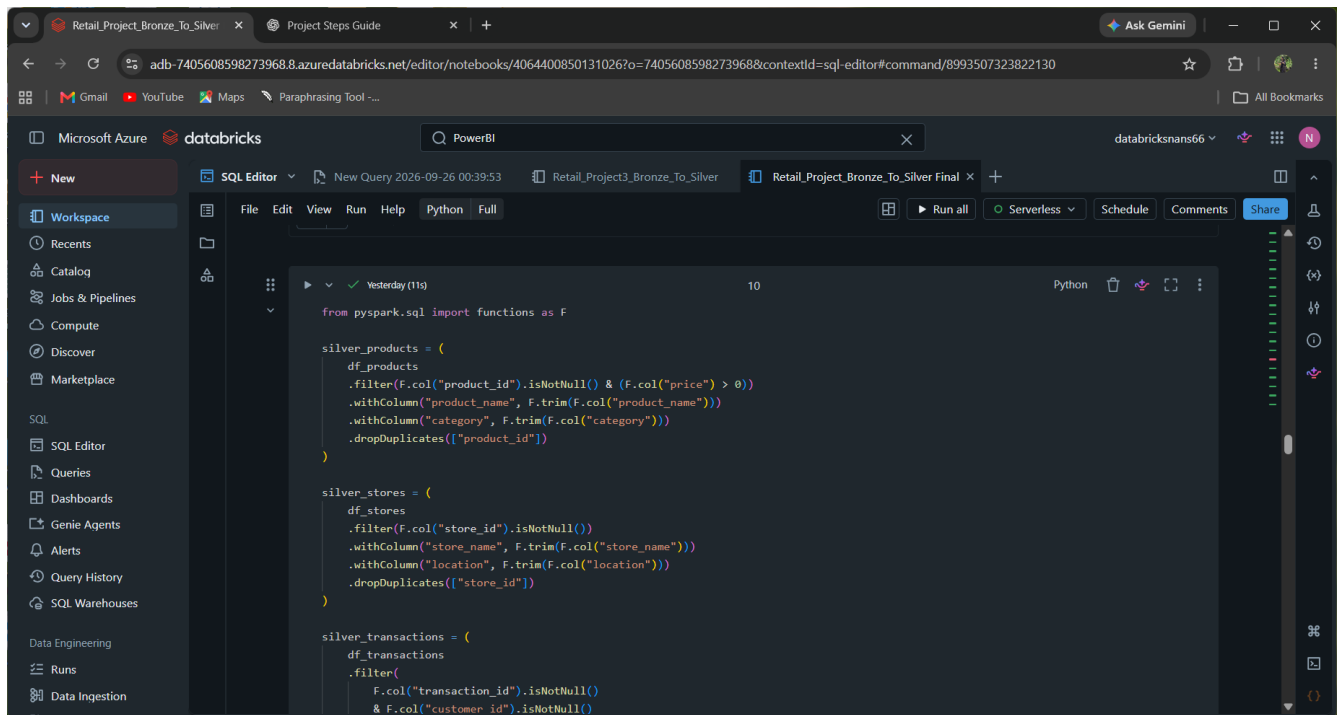


Figure 22. Silver cleaning logic

Visible code filters invalid product rows, trims text and removes duplicate product IDs.

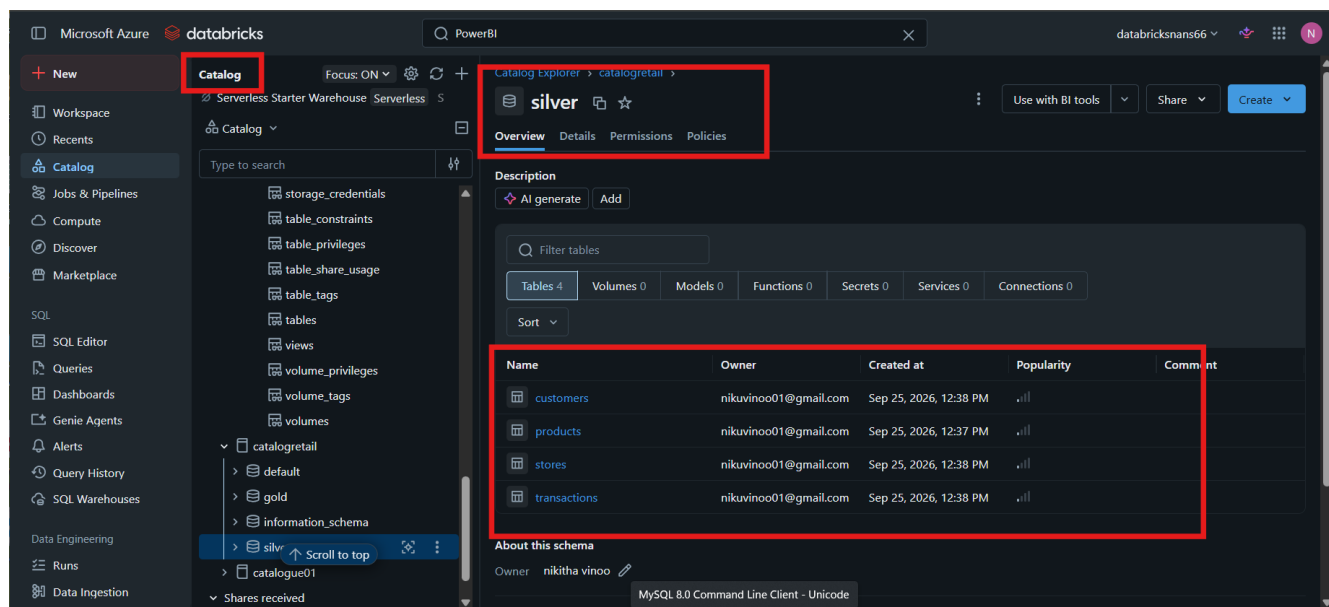


Figure 23. Silver tables in Catalog Explorer

The catalog lists customers, products, stores and transactions.

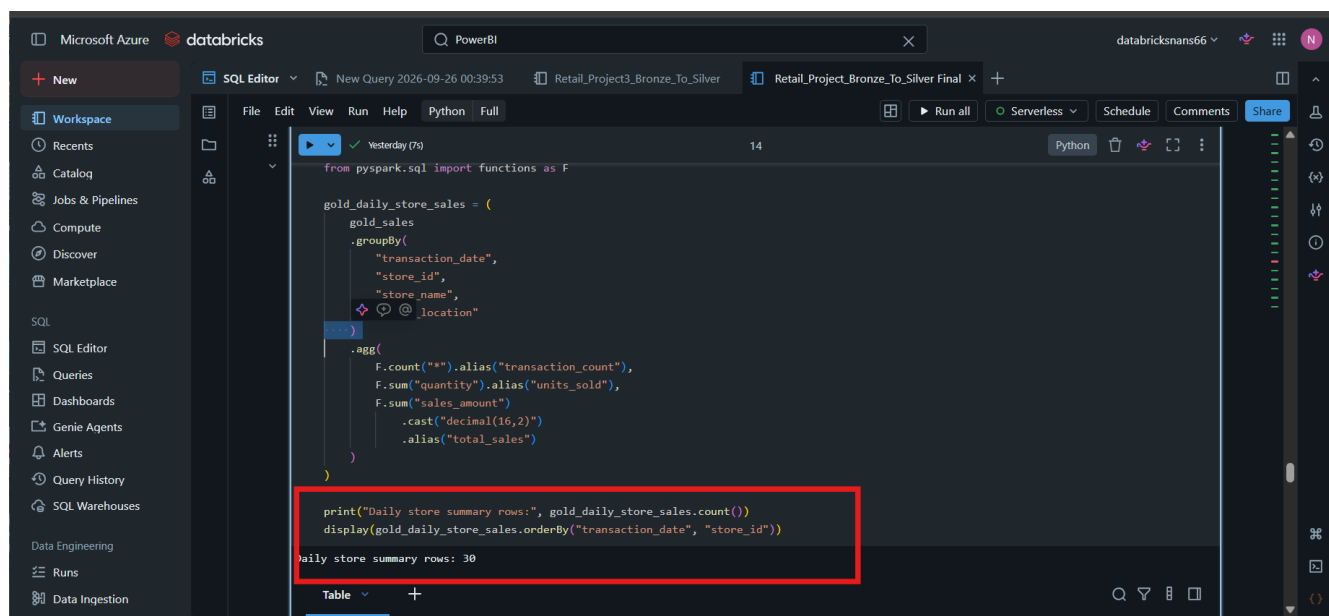


Figure 24. Daily store sales aggregation

The notebook groups gold sales by date and store and aggregates transaction count, units and sales.

Daily store summary rows: 30

	transaction_date	store_id	store_name	store_location	transaction_count	units_sold	total_sales
1	2024-07-10	5	Mega Plaza	Chennai	1	5	1995.0
2	2024-07-14	4	Downtown Mini Store	Pune	1	4	1596.0
3	2024-07-14	5	Mega Plaza	Chennai	1	1	799.9
4	2024-07-17	4	Downtown Mini Store	Pune	1	5	3999.9
5	2024-07-30	1	City Mall Store	Mumbai	1	2	298.0
6	2024-08-11	3	Tech World Outlet	Bangalore	1	2	1599.9
7	2024-08-27	2	High Street Store	Delhi	1	5	6497.4
8	2024-09-05	4	Downtown Mini Store	Pune	1	3	2399.9
9	2024-09-21	2	High Street Store	Delhi	1	2	598.0
10	2024-10-02	3	Tech World Outlet	Bangalore	1	5	3999.9
11	2024-10-08	3	Tech World Outlet	Bangalore	1	2	1599.9
12	2024-10-10	1	City Mall Store	Mumbai	1	1	399.0
13	2024-11-02	1	City Mall Store	Mumbai	1	1	399.0
14	2024-11-12	4	Downtown Mini Store	Pune	1	5	2495.0

30 rows · 7.287s runtime · Refreshed Yesterday

Figure 25. Daily store sales output

A table preview shows date, store, location, transaction count, units sold and total sales.

```
display(
  spark.read.table("catalogretail.gold.daily_store_sales")
    .orderBy("transaction_date", "store_id")
)
```

	transaction_date	store_id	store_name	store_location	transaction_count	units_sold	total_sales
1	2024-07-10	5	Mega Plaza	Chennai	1	5	1995.0
2	2024-07-14	4	Downtown Mini Store	Pune	1	4	1596.0
3	2024-07-14	5	Mega Plaza	Chennai	1	1	799.9
4	2024-07-17	4	Downtown Mini Store	Pune	1	5	3999.9
5	2024-07-30	1	City Mall Store	Mumbai	1	2	298.0
6	2024-08-11	3	Tech World Outlet	Bangalore	1	2	1599.9
7	2024-08-27	2	High Street Store	Delhi	1	5	6497.4
8	2024-09-05	4	Downtown Mini Store	Pune	1	3	2399.9
9	2024-09-21	2	High Street Store	Delhi	1	2	598.0
10	2024-10-02	3	Tech World Outlet	Bangalore	1	5	3999.9
11	2024-10-08	3	Tech World Outlet	Bangalore	1	2	1599.9
12	2024-10-10	1	City Mall Store	Mumbai	1	1	399.0

Figure 26. Reading the saved gold table

The notebook queries catalogretail.gold.daily_store_sales.

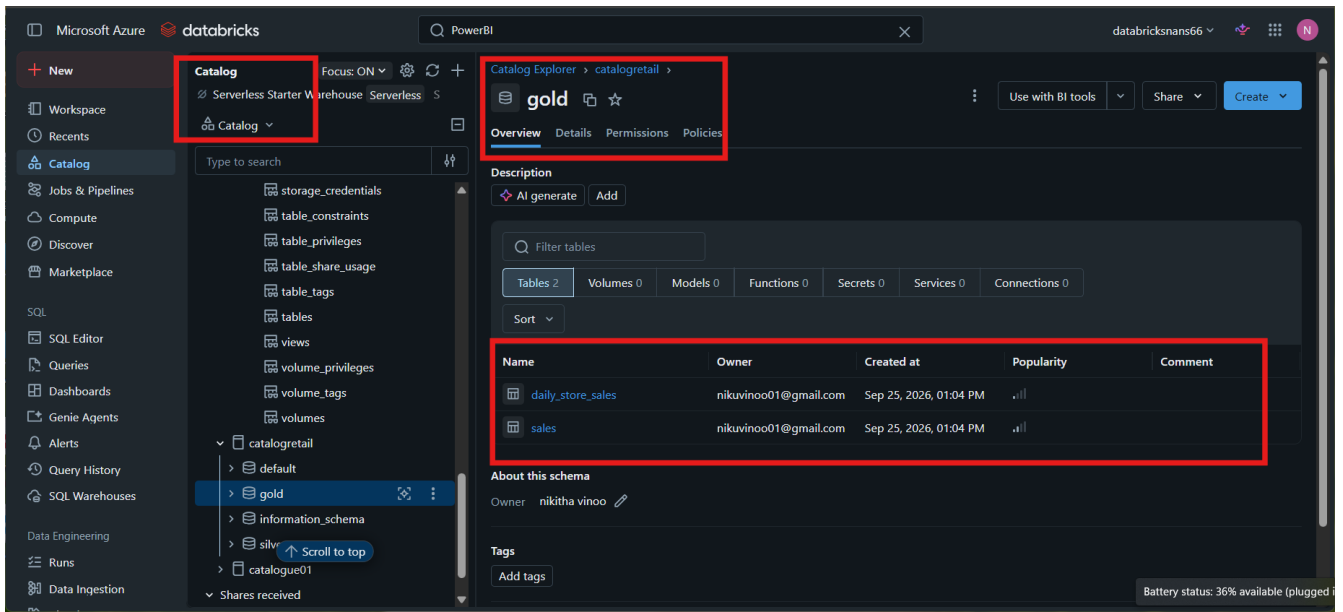


Figure 27. Gold tables in Catalog Explorer

The gold schema lists daily_store_sales and sales.

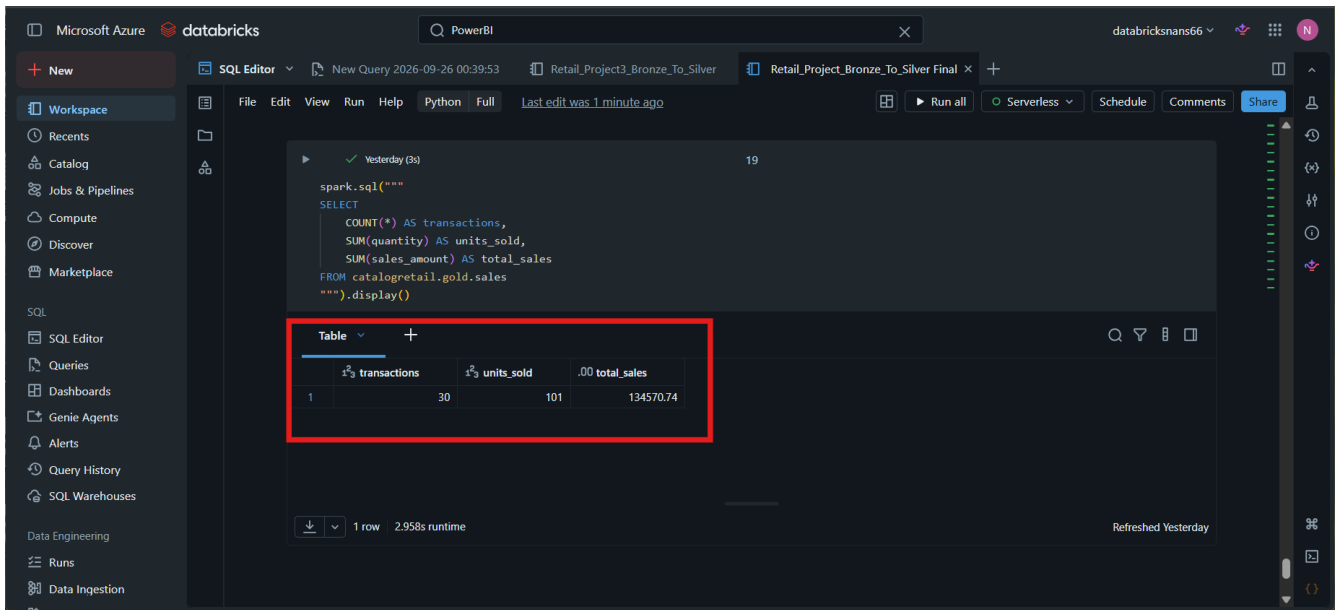


Figure 28. Gold sales validation query

The captured result is 30 transactions, 101 units sold and 134,570.74 total sales.

6 Power BI connection and dashboard

Power BI connects to the Databricks SQL warehouse and imports the analytical tables. The report uses sales amounts and store, category and date fields to present retail results. A card summarizes sales, while other visuals show store comparison and trends. The screenshots below include setup steps as well as the completed report.

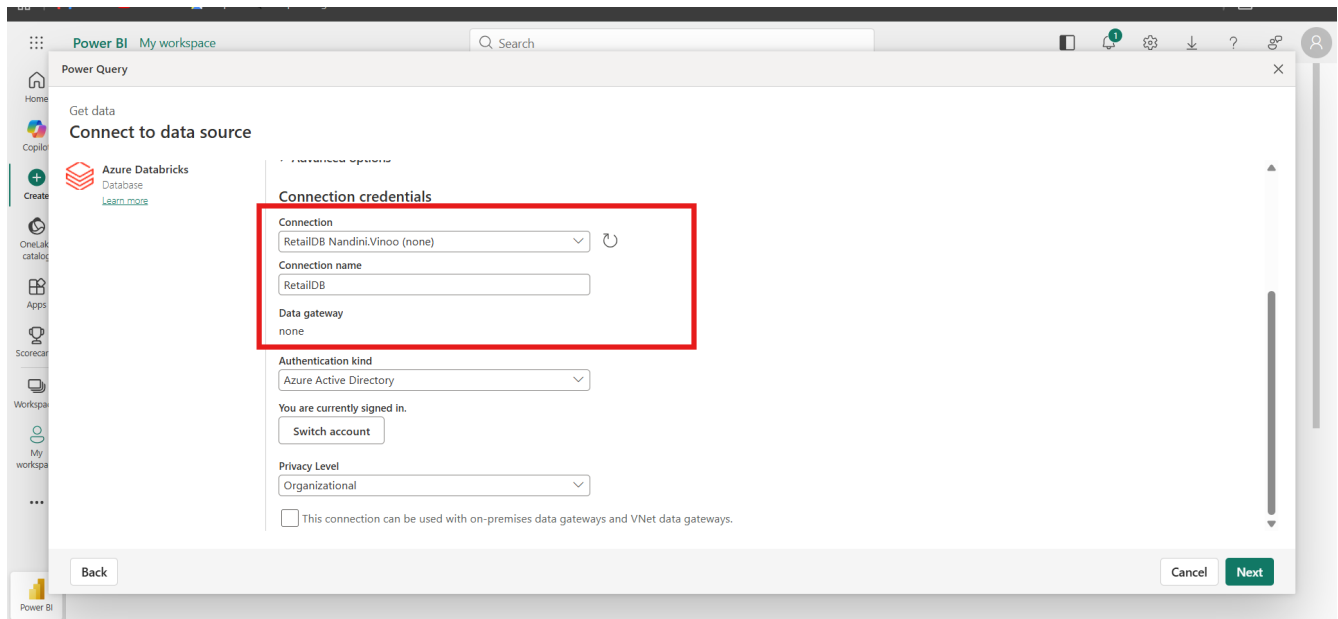


Figure 29. Power BI Databricks connection credentials

The connector sign in screen is shown during setup.

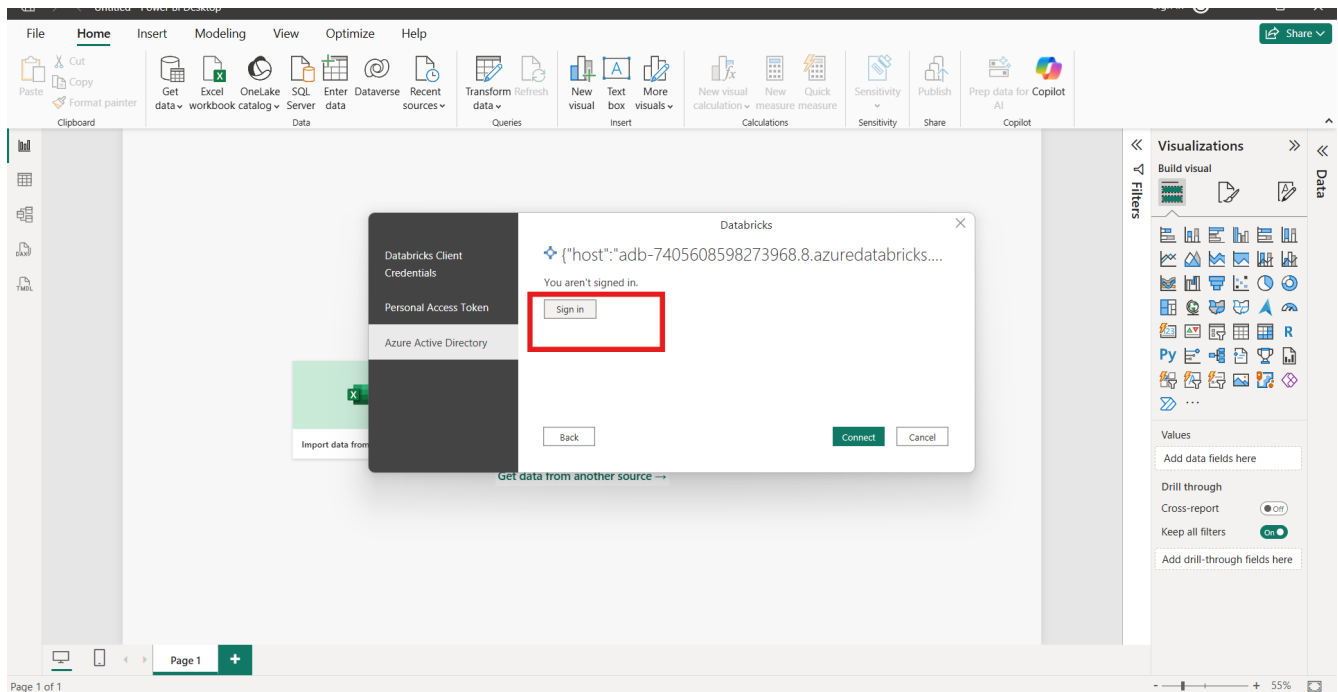


Figure 30. Connecting Power BI to the warehouse

The Databricks host and HTTP path connection dialog is visible.

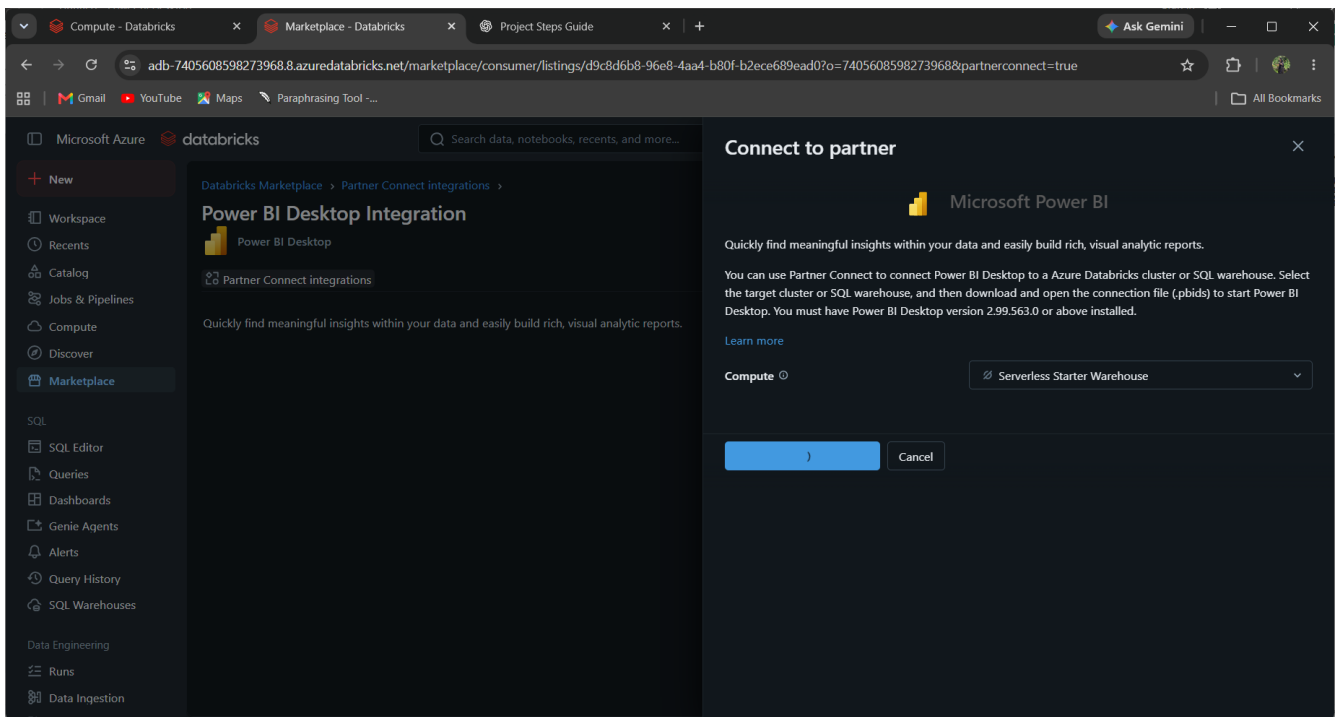


Figure 31. Databricks connection details

The warehouse is selected for BI tool access.

Navigator

Display Options

adb-7405608598273968.8.azuredatabricks.net:...

catalogretail [3]

default

gold [2]

☒ daily_store_sales

☐ sales

silver

catalogue01

databricksnans66_7405608598273968

samples

system

daily_store_sales

transaction_date	store_id	store_name	store_location	transac
10-10-2024	1	City Mall Store	Mumbai	
17-07-2024	4	Downtown Mini Store	Pune	
14-07-2024	5	Mega Plaza	Chennai	
17-03-2025	2	High Street Store	Delhi	
29-11-2024	2	High Street Store	Delhi	
05-09-2024	4	Downtown Mini Store	Pune	
10-07-2024	5	Mega Plaza	Chennai	
13-12-2024	4	Downtown Mini Store	Pune	
30-07-2024	1	City Mall Store	Mumbai	
04-01-2025	3	Tech World Outlet	Bangalore	
02-10-2024	3	Tech World Outlet	Bangalore	
21-09-2024	2	High Street Store	Delhi	
31-03-2025	4	Downtown Mini Store	Pune	
16-11-2024	3	Tech World Outlet	Bangalore	
16-03-2025	4	Downtown Mini Store	Pune	
08-06-2025	5	Mega Plaza	Chennai	
30-04-2025	4	Downtown Mini Store	Pune	
19-05-2025	5	Mega Plaza	Chennai	
04-05-2025	3	Tech World Outlet	Bangalore	
08-10-2024	3	Tech World Outlet	Bangalore	
12-11-2024	4	Downtown Mini Store	Pune	
27-08-2024	2	High Street Store	Delhi	
14-07-2024	4	Downtown Mini Store	Pune	

Load

Transform Data

Cancel

Figure 32. Navigator table selection

The daily store sales table is selected for loading.

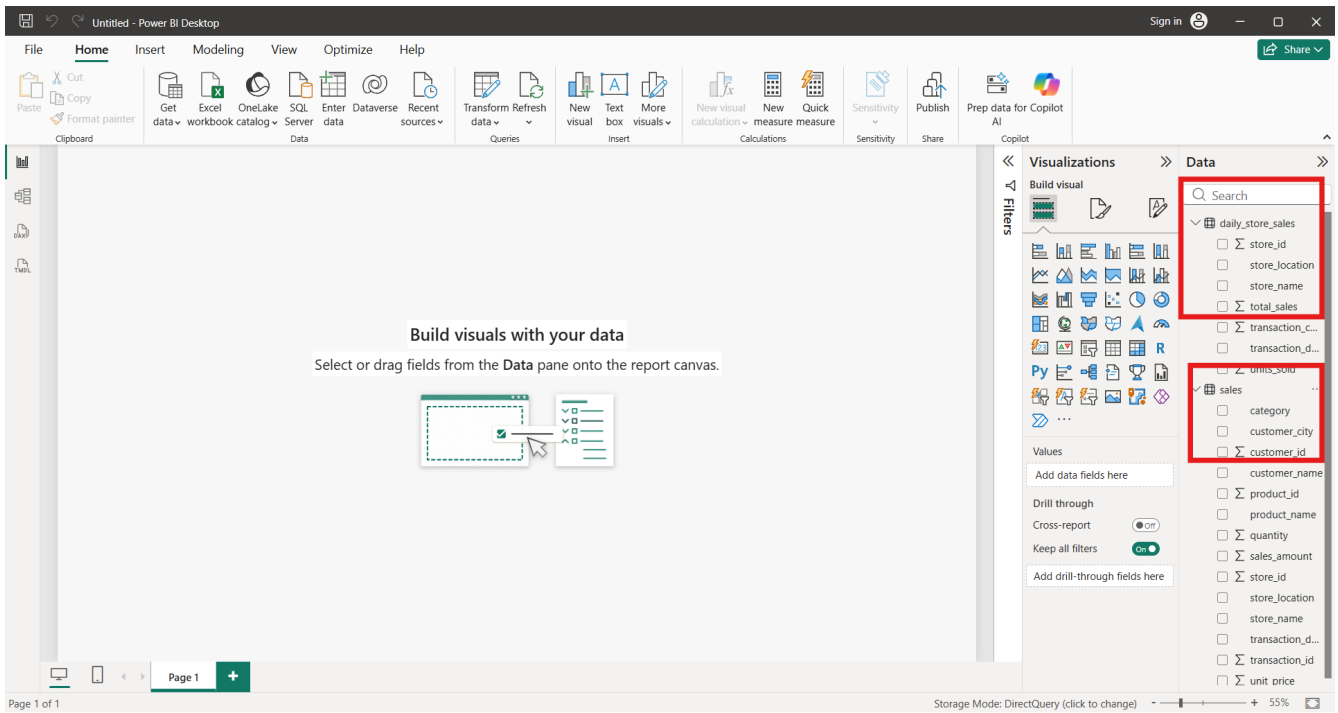


Figure 33. Report field list

The first visual is assembled using the imported table fields.

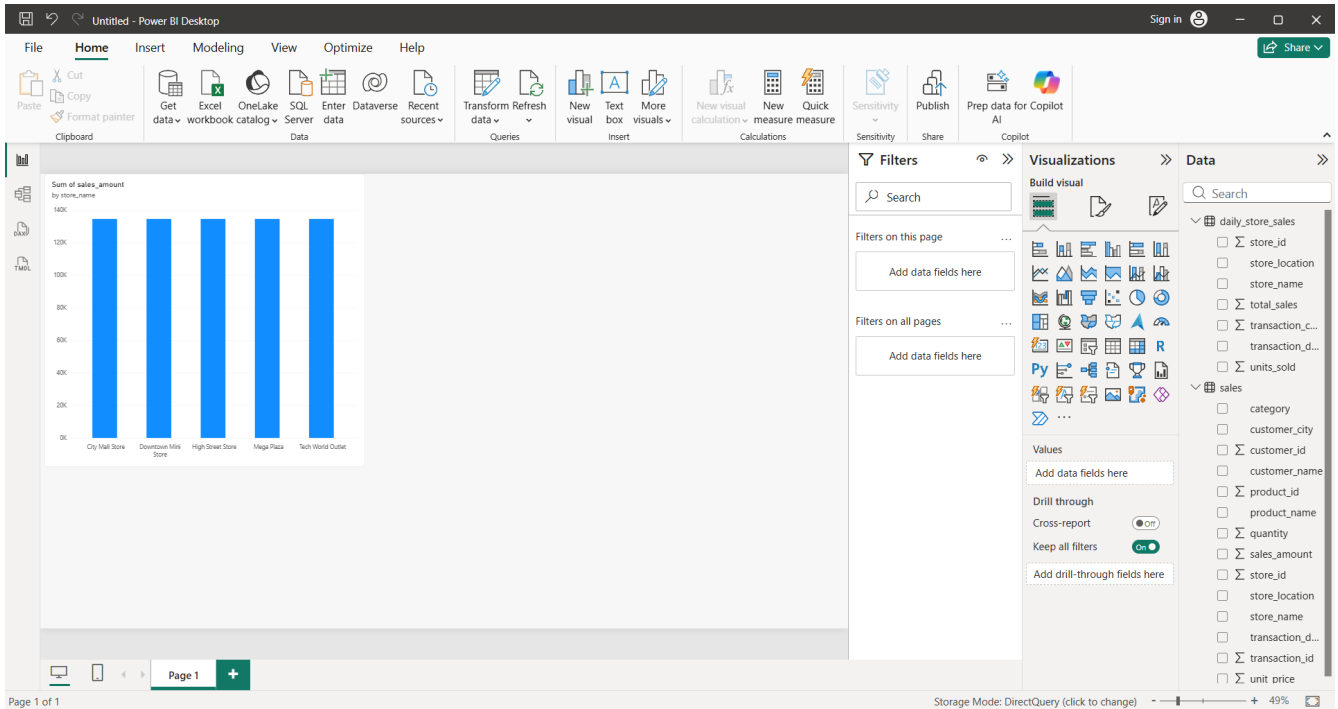


Figure 34. Initial store chart

The first column chart appears on the report canvas.

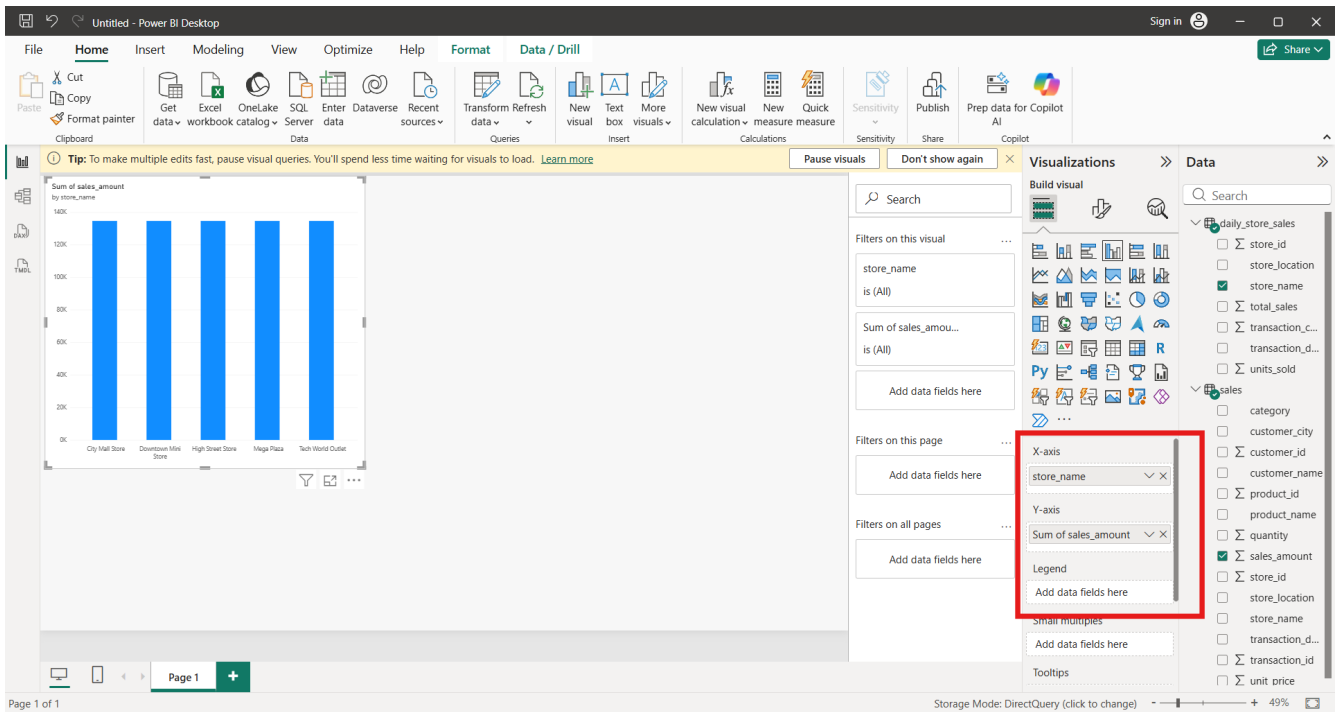


Figure 35. Configuring a second visual

The visual fields are set from the loaded data.

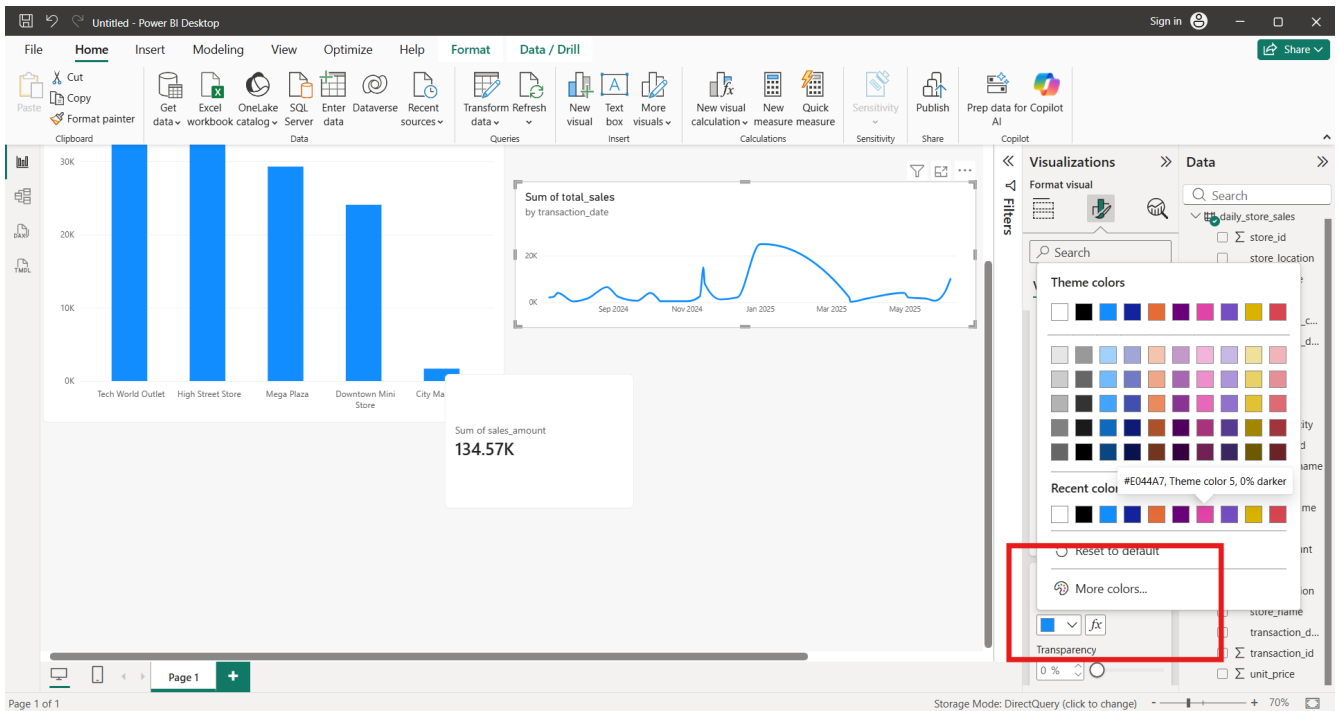


Figure 36. Sales trend and KPI visuals

The report shows a line trend and a sales card alongside the bars.

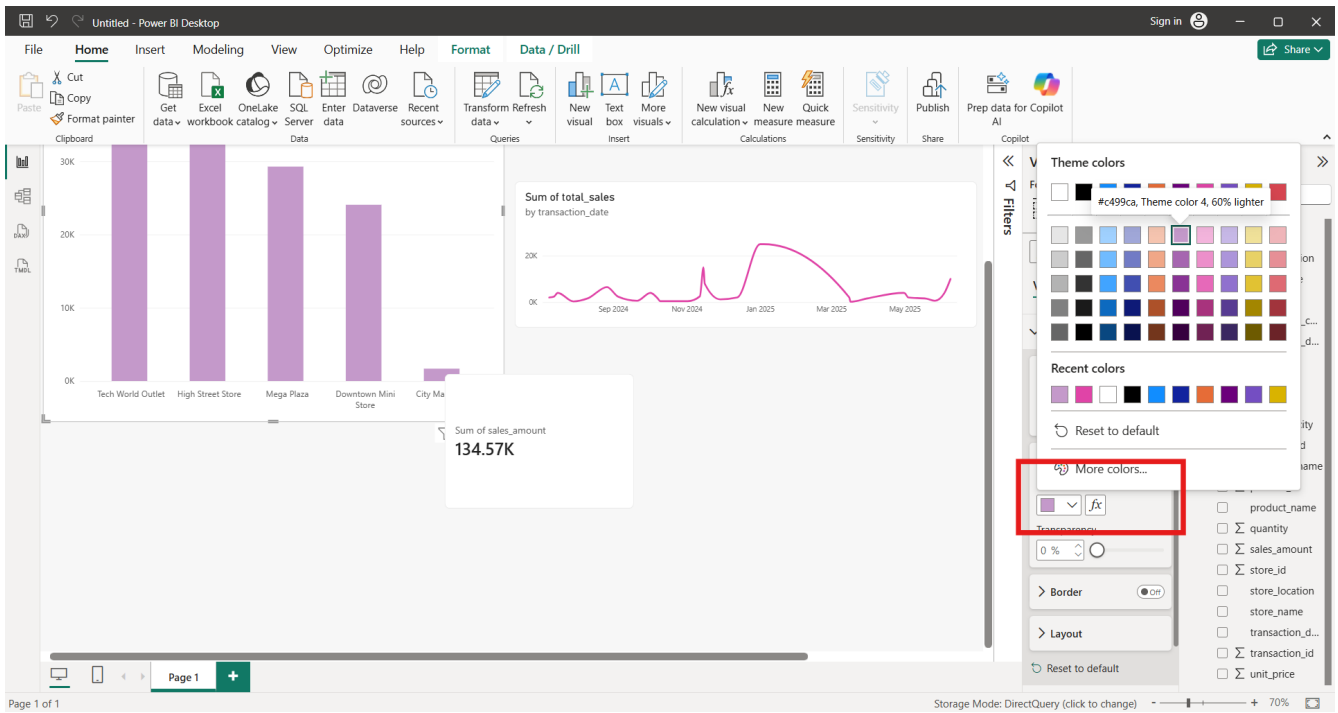


Figure 37. Changing chart color

Formatting options are used to adjust the chart appearance.

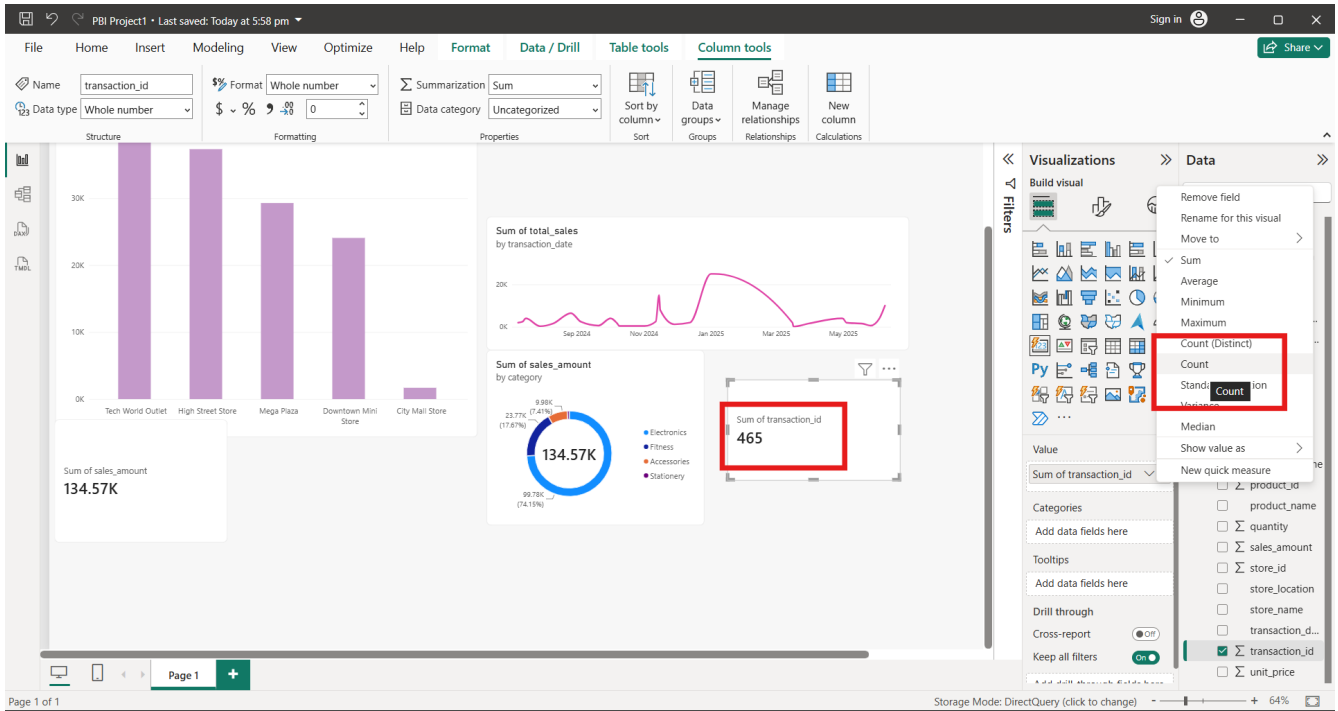


Figure 38. Dashboard measure selection

The report displays sales by store, category, trend and KPI values.

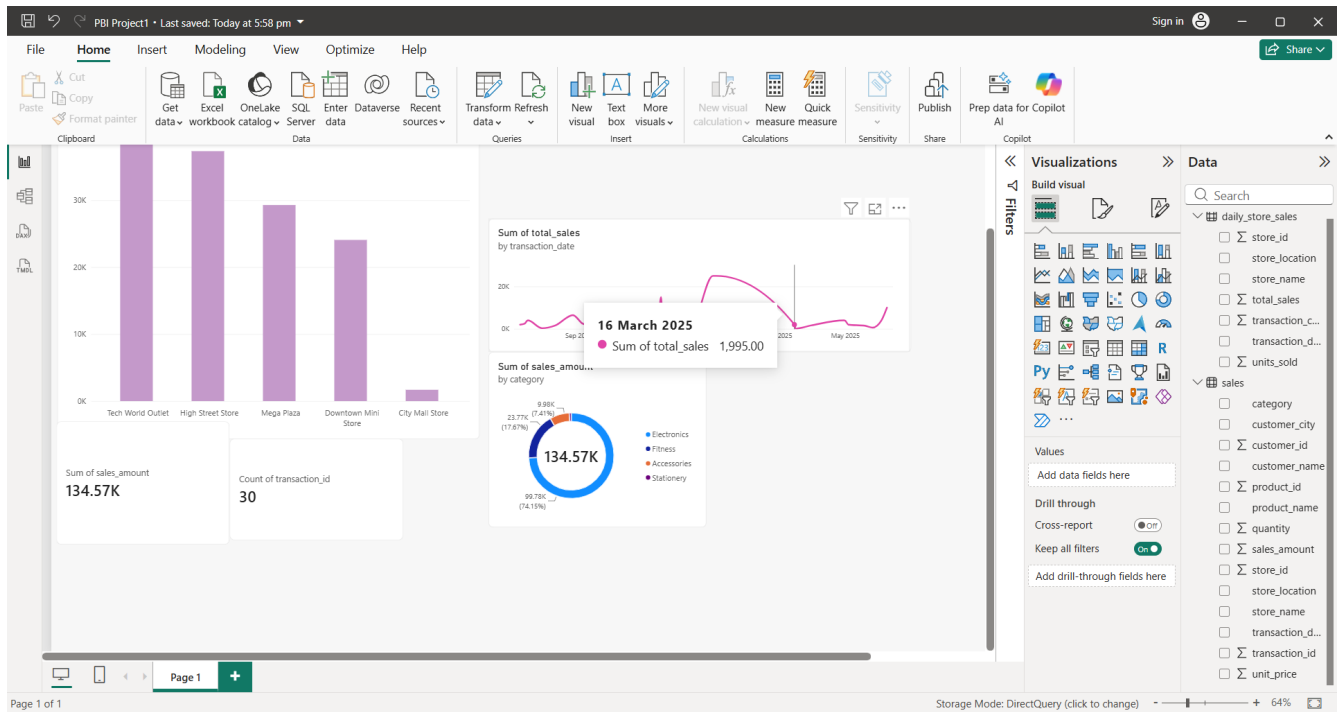


Figure 39. Additional dashboard visual

The report layout is refined with the chosen fields.

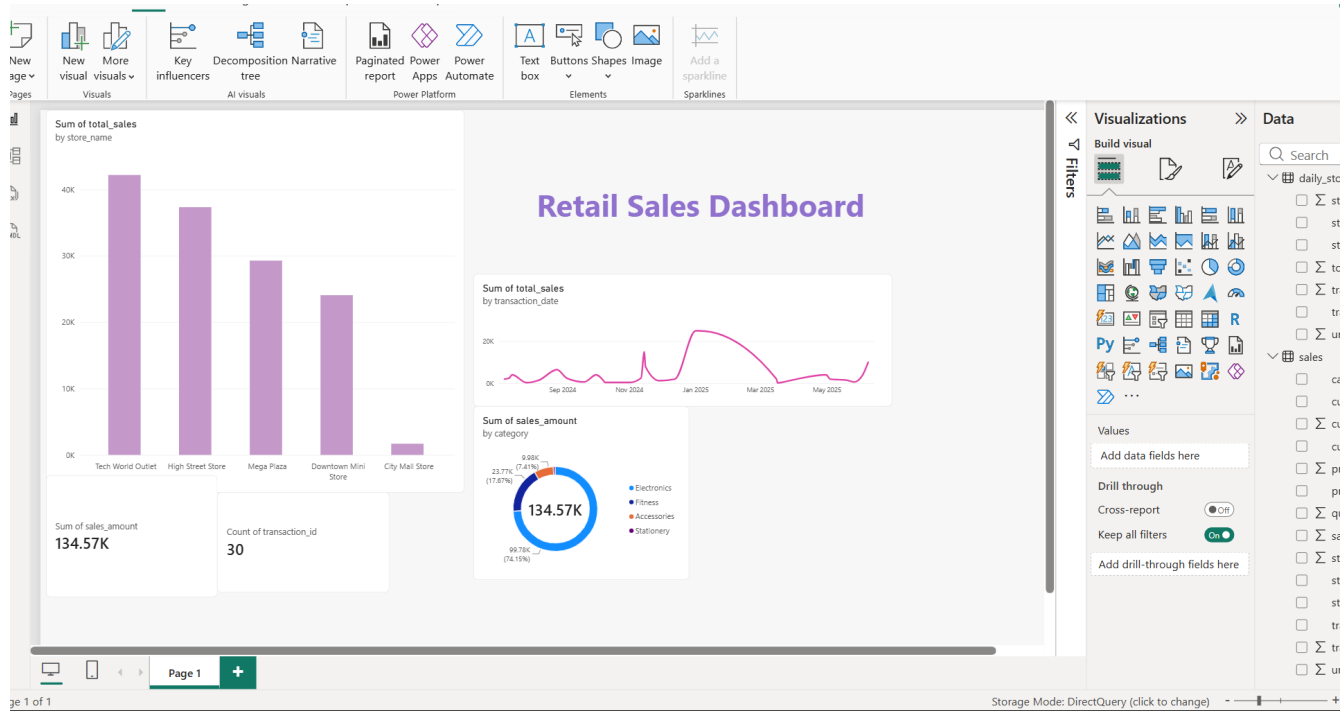


Figure 40. Dashboard title added

The title identifies the page as Retail Sales Dashboard.

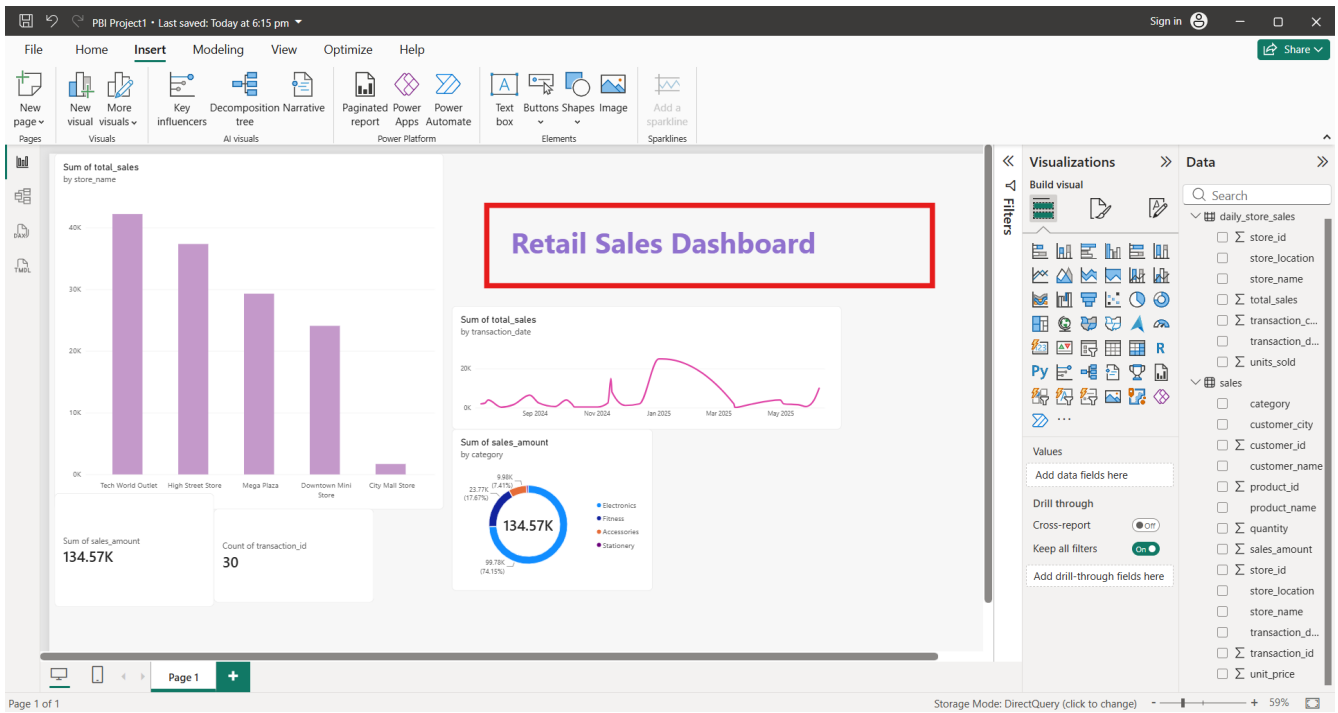


Figure 41. Title formatting

The text box and report visuals are shown together.

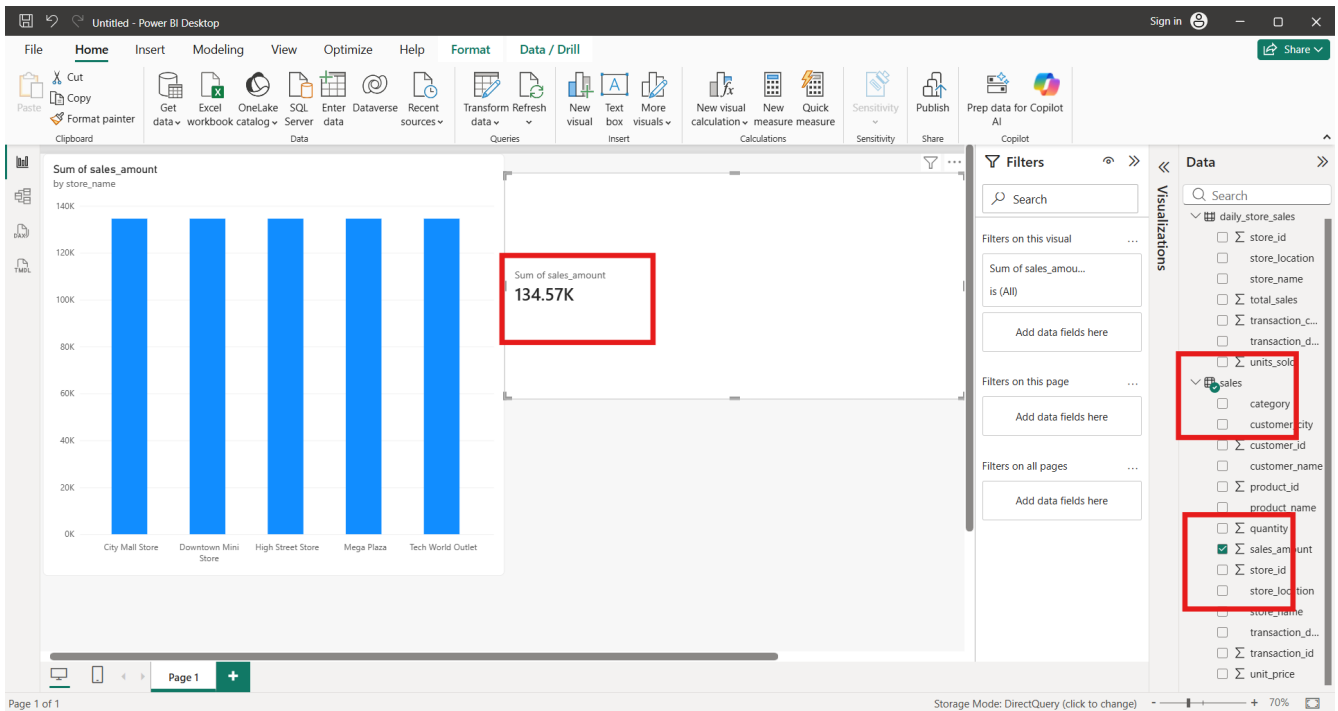


Figure 42. Sales card closeup

The card shows approximately 134.57K in total sales.

7 Project outcome and handoff

The documented flow starts with three Azure SQL tables and the customers HTTP feed, lands data through ADF into ADLS bronze, produces silver and gold datasets in Databricks, and presents the result in Power BI. The captured Databricks check confirms 30 transactions, 101 units sold and total sales of 134,570.74. The published ADF pipeline, catalog tables, storage containers and dashboard are evidenced by the supplied screenshots.

For a future rerun, check the current SQL schema against the reconstructed SQL example, validate the customer endpoint and linked services, execute the ADF pipeline, run the Databricks notebook, confirm the gold totals, and refresh Power BI. This is a handoff sequence; the screenshots do not prove that an automated recurring schedule is configured.

Screenshot provenance

Figures in this report are taken from the three supplied Power BI Project Screenshots PDF files and the corresponding uploaded screenshot images. The SQL code panel is explicitly labeled as a reconstruction because the original SQL table creation capture was unavailable.